

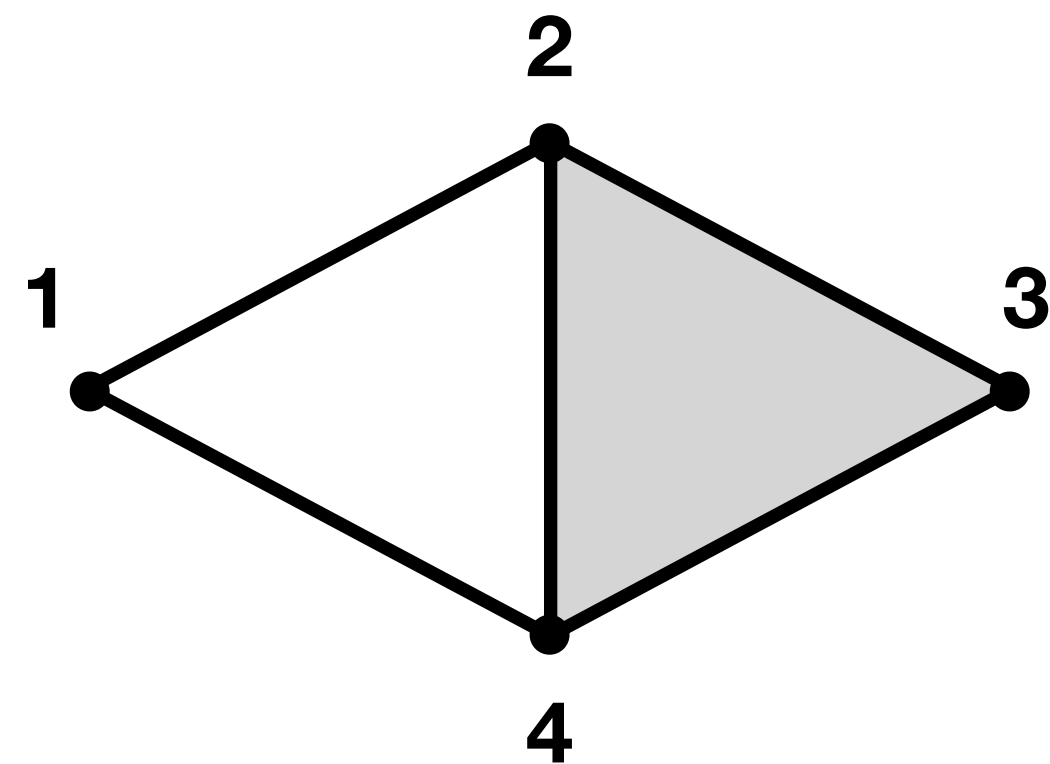
Topological Data Analysis

Lecture 10

Computational persistent homology

Oleg Kachan

Boundary matrix

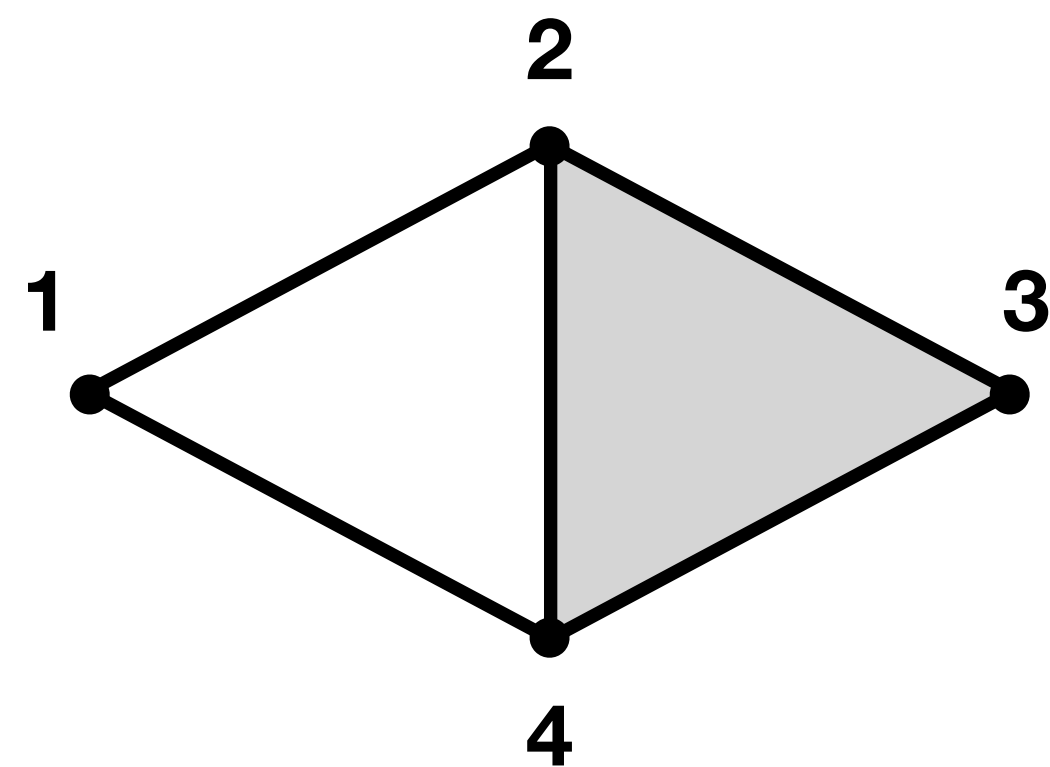


B =

	1	2	3	4	12	14	23	24	34	234
0	B ₀									
1					B ₁					
2										
3										
4										
12										B ₂
14										
23										
24										
34										

$$K = \{\{0\}, \{1\}, \{2\}, \{3\}, \{12\}, \{14\}, \{23\}, \{24\}, \{34\}, \{234\}\}$$

Boundary matrix

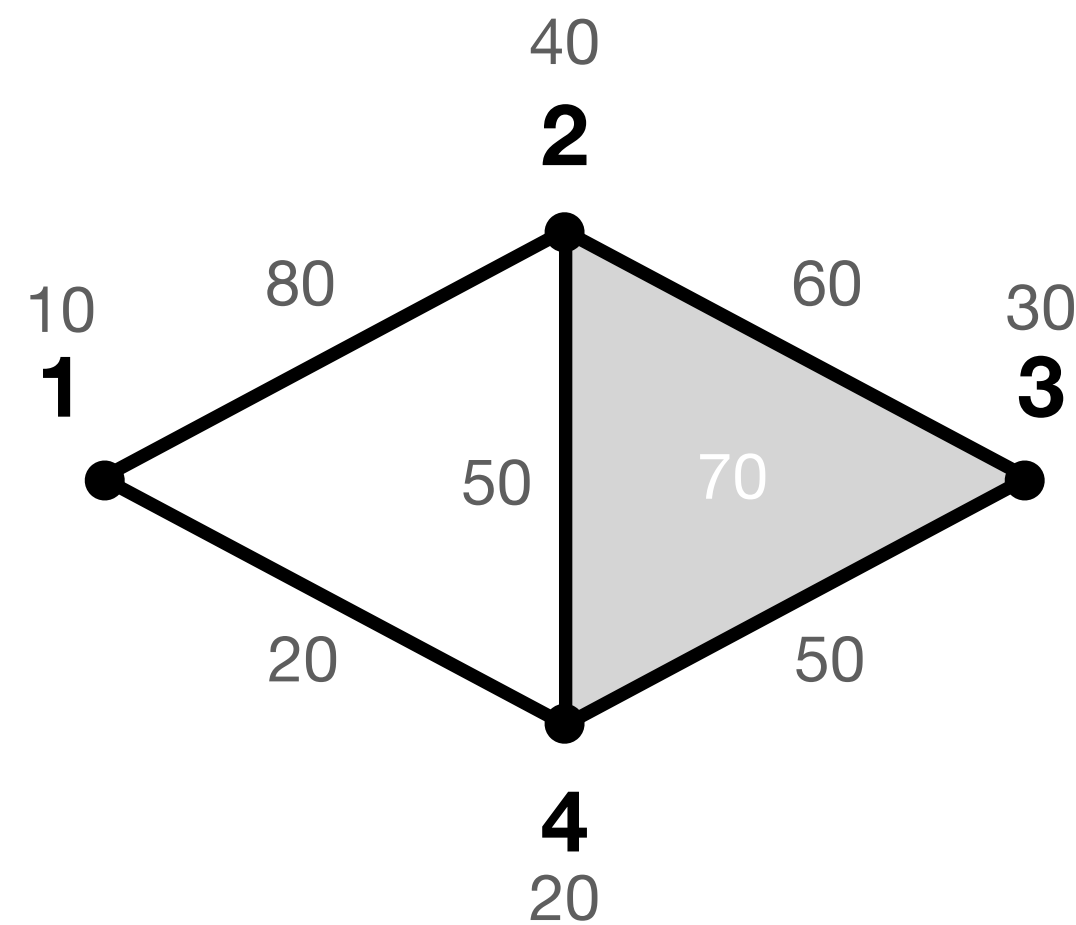


B =

	1	2	3	4	12	14	23	24	34	234
0	1	1	1	1						
1					1	1				
2					1		1	1		
3							1		1	
4						1		1	1	
12										
14										
23										
24										
34										

One may skip adding **B**₀ to the full matrix or zero out it

Filtration function



A function $f : K \rightarrow \mathbb{R}$ is called a filtration function iff, either

$$f(\tau) \leq f(\sigma) \iff \tau \subseteq \sigma \quad \text{(sublevel filtration)}$$

$$f(\tau) \geq f(\sigma) \iff \tau \supseteq \sigma \quad \text{(superlevel filtration)}$$

$$K_t = \{\sigma \in K \mid f(\sigma) \leq t\}$$

sublevel set

$$t \in (-\infty, +\infty)$$

$$K^t = \{\sigma \in K \mid f(\sigma) \geq t\}$$

superlevel set

$$t \in (+\infty, -\infty)$$

A filtration is a sequence of sublevel (superlevel) sets s.t.

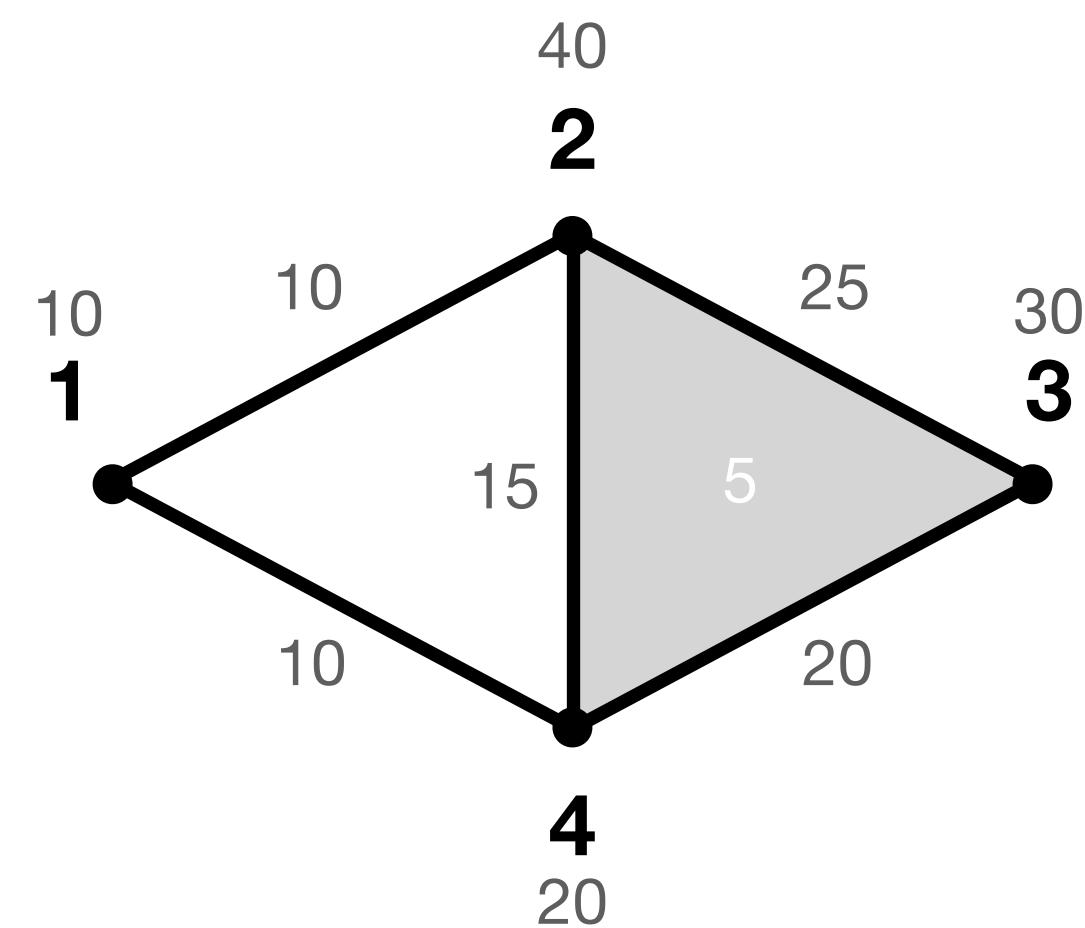
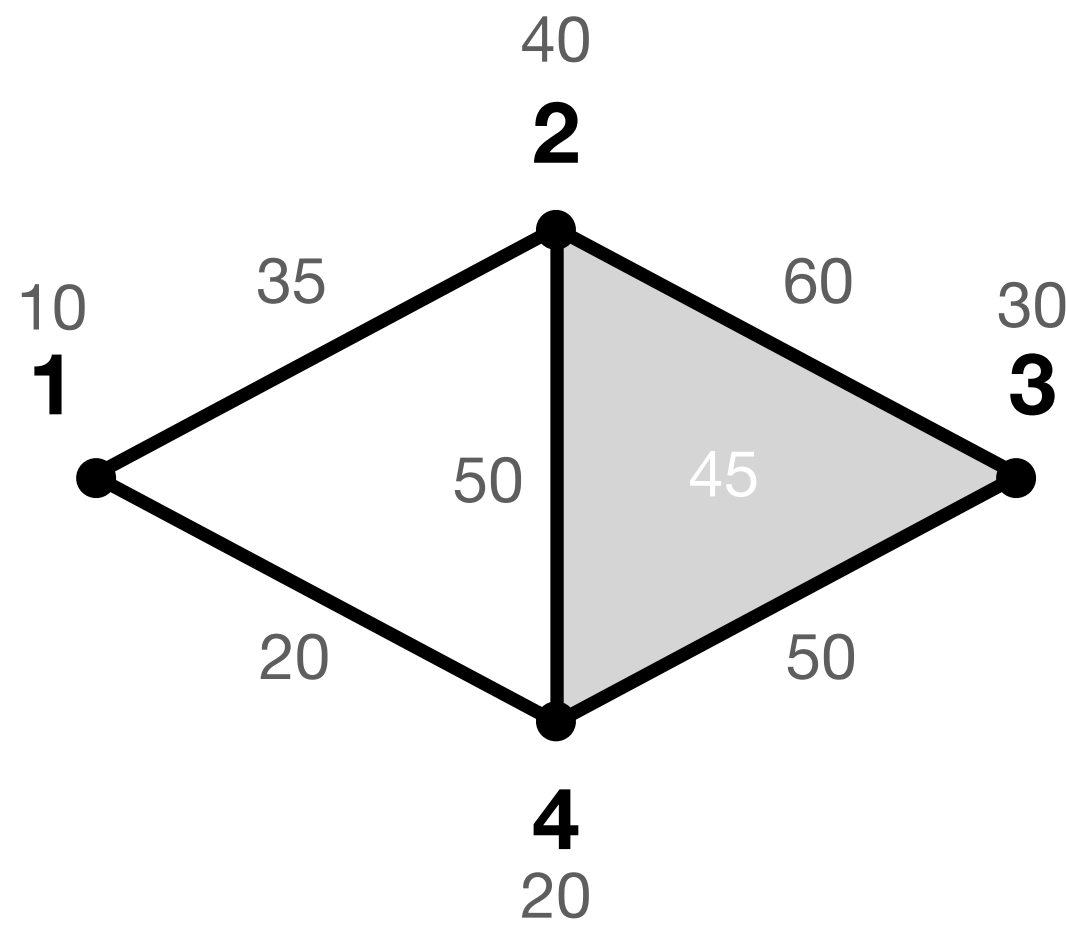
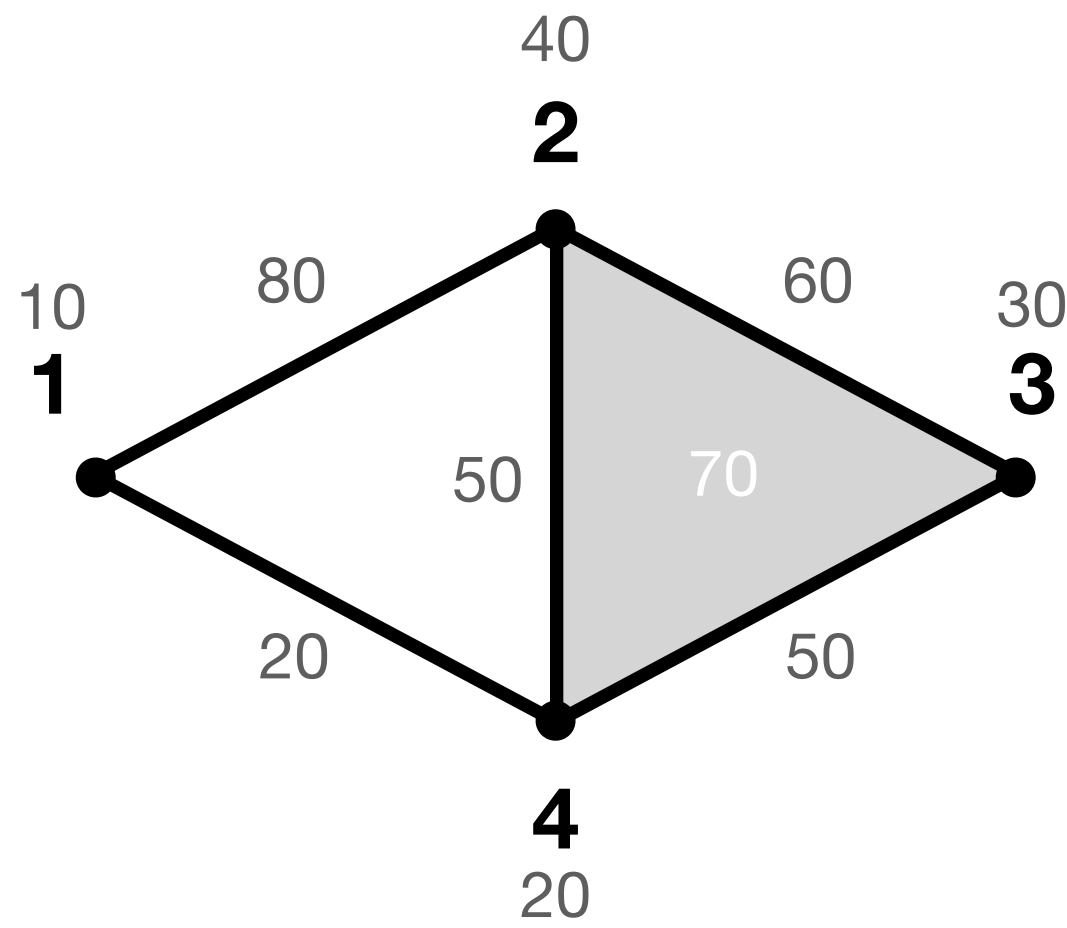
$$\emptyset \subset K_1 \subset K_2 \subset \dots \subset K$$

sublevel filtration

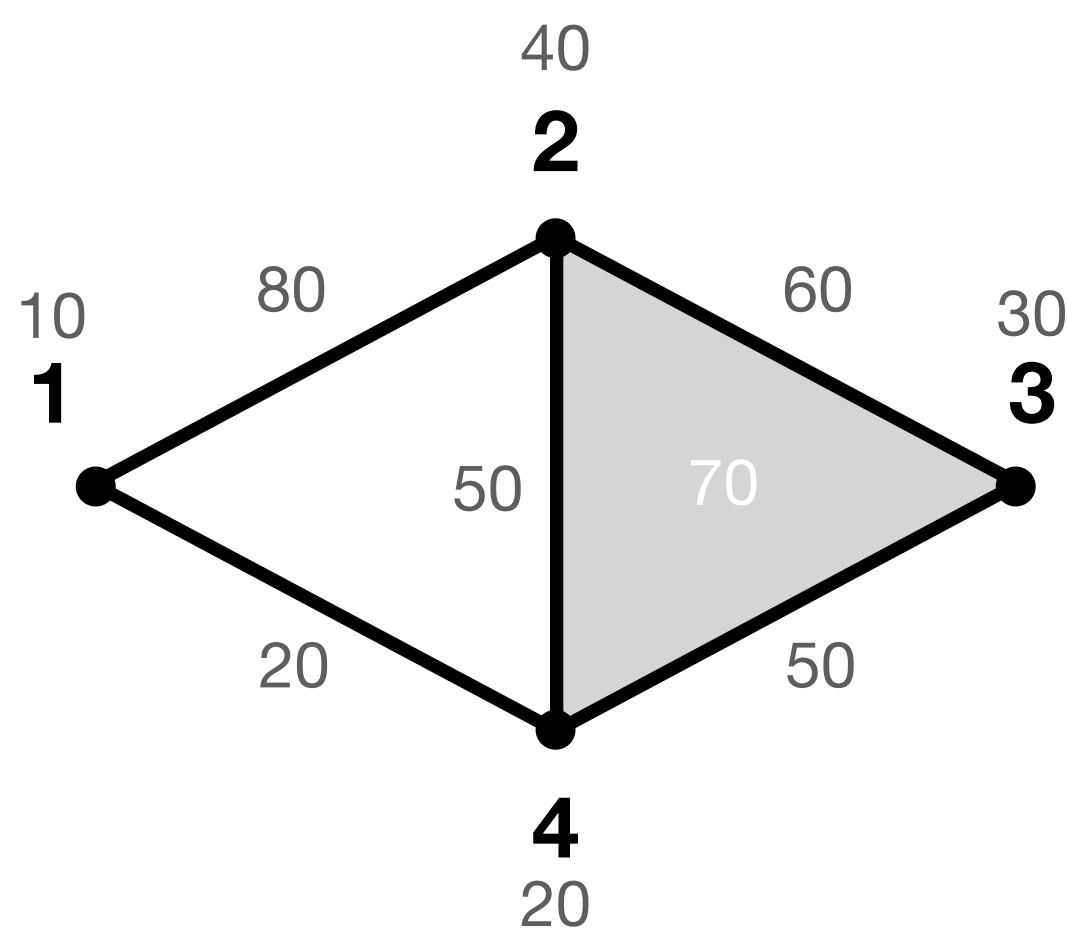
$$\emptyset \subset K^1 \subset K^2 \subset \dots \subset K$$

superlevel filtration

Filtration function



Filtered simplicial complex



B =

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
0										
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

Compatible ordering

Filtration function provides order on simplices, therefore on columns and rows of the filtration matrix. Ties are broken, first by simplex dimension, second by lexicographic order given by order on vertices.

Standard algorithm for computing persistent homology

Given a $n \times n$ matrix ∂ define a function mapping a column to the row index of its lowest nonzero element

$$\text{low}(j) = \begin{cases} \max\{i \mid \partial_{ij} \neq 0\}, & \partial_j \neq 0, \\ -1, & \text{otherwise.} \end{cases}$$

Standard algorithm

- reduce ∂ by column additions
- select a current column ∂_j , moving from left to right
- for each $k < j$ add columns ∂_k to current column ∂_j if $\text{low}(k) = \text{low}(j)$
- matrix is reduced when $\text{low}(\cdot)$ is injective

Algorithm 3: Standard algorithm over $\mathbb{F}_2$

Input: An $n \times n$ boundary matrix ∂ over $\mathbb{F}_2$

```
1 for  $j = 0, \dots, n - 1$  do
2   |   while  $\exists k < j$  such that  $\text{low}_\partial(k) = \text{low}_\partial(j) > -1$  do
3   |   |    $\partial_j \leftarrow \partial_j + \partial_k$ 
4   |   end while
5 end for
6 return  $\partial$ 
```

Boundary matrix reduction

10
1
●

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=10

Boundary matrix reduction

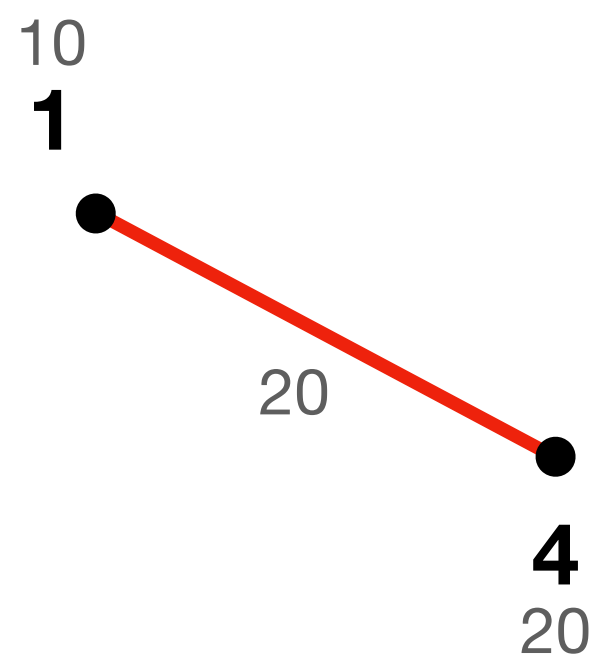
10
1
●

●
4
20

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=20

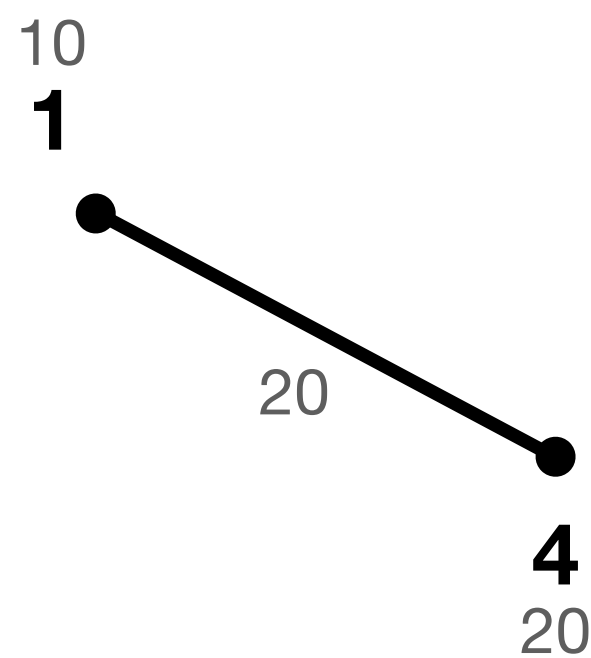
Boundary matrix reduction



	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=20

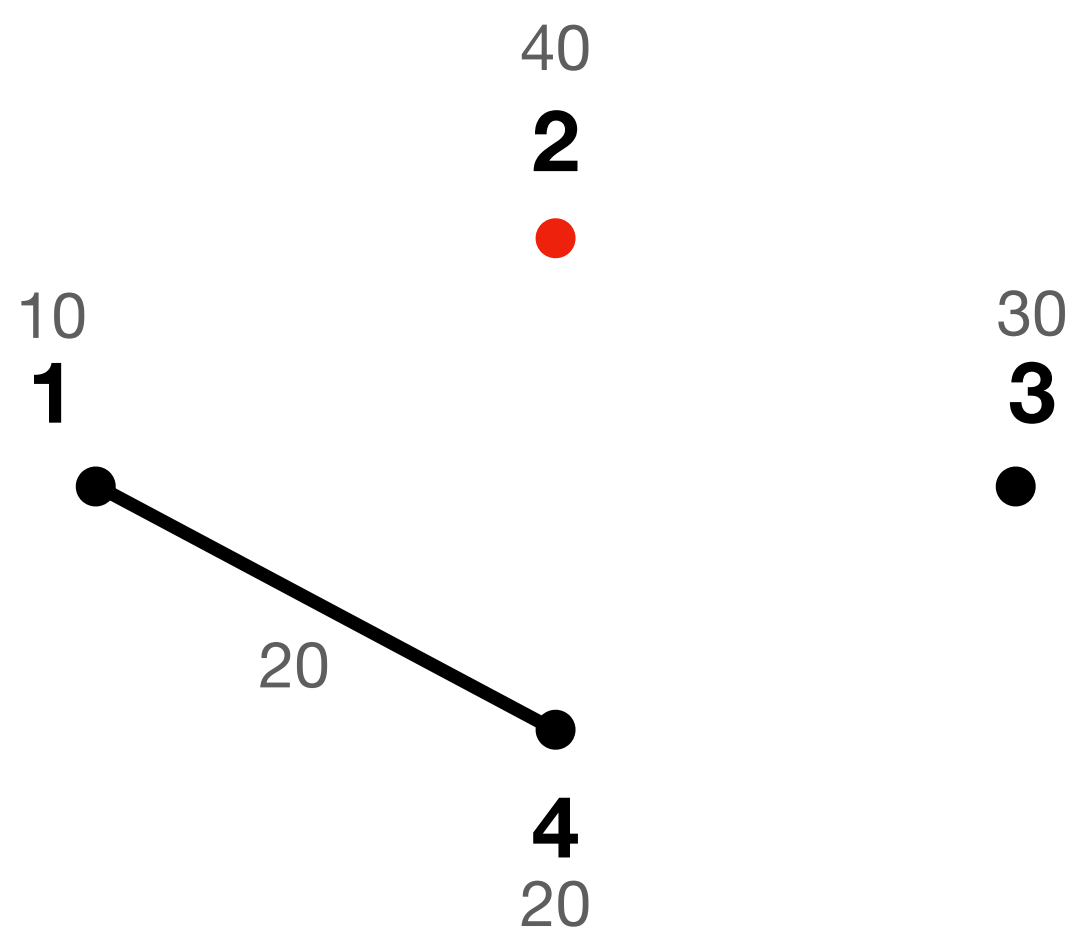
Boundary matrix reduction



	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=30

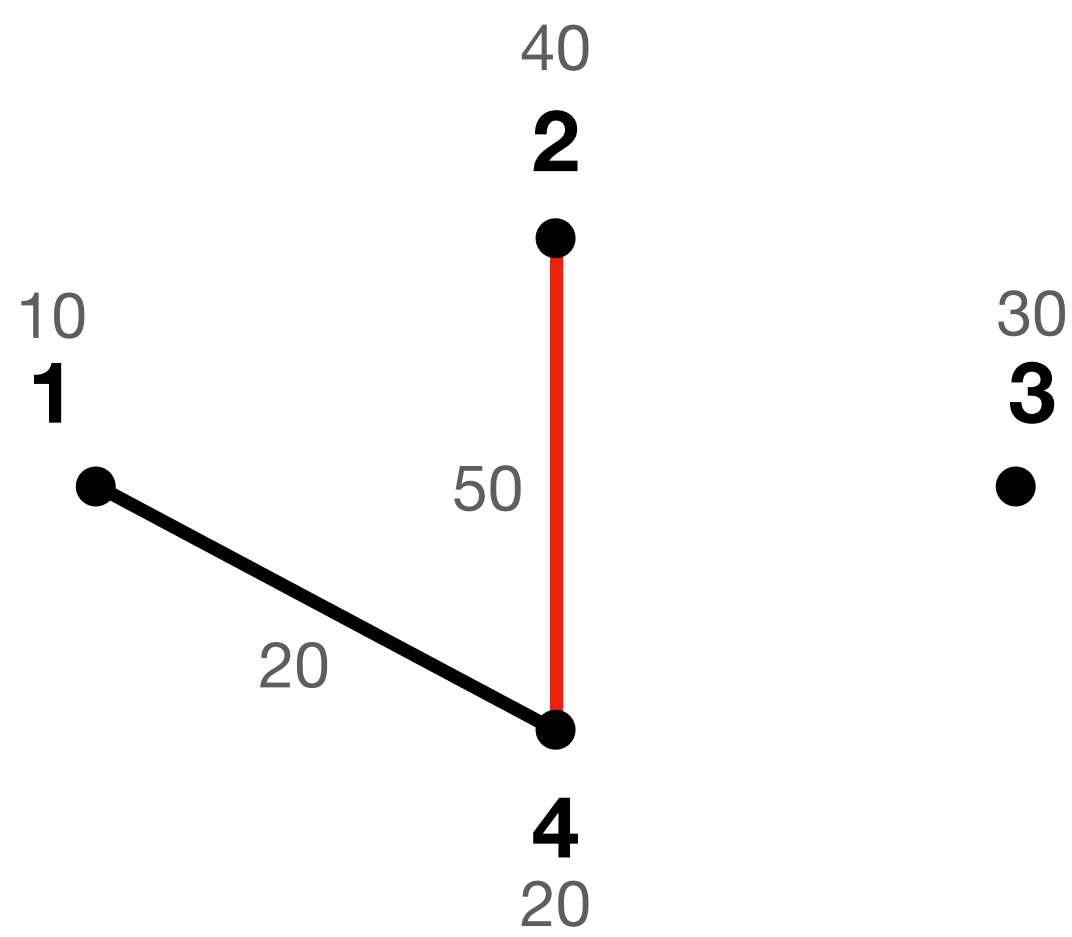
Boundary matrix reduction



	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=40

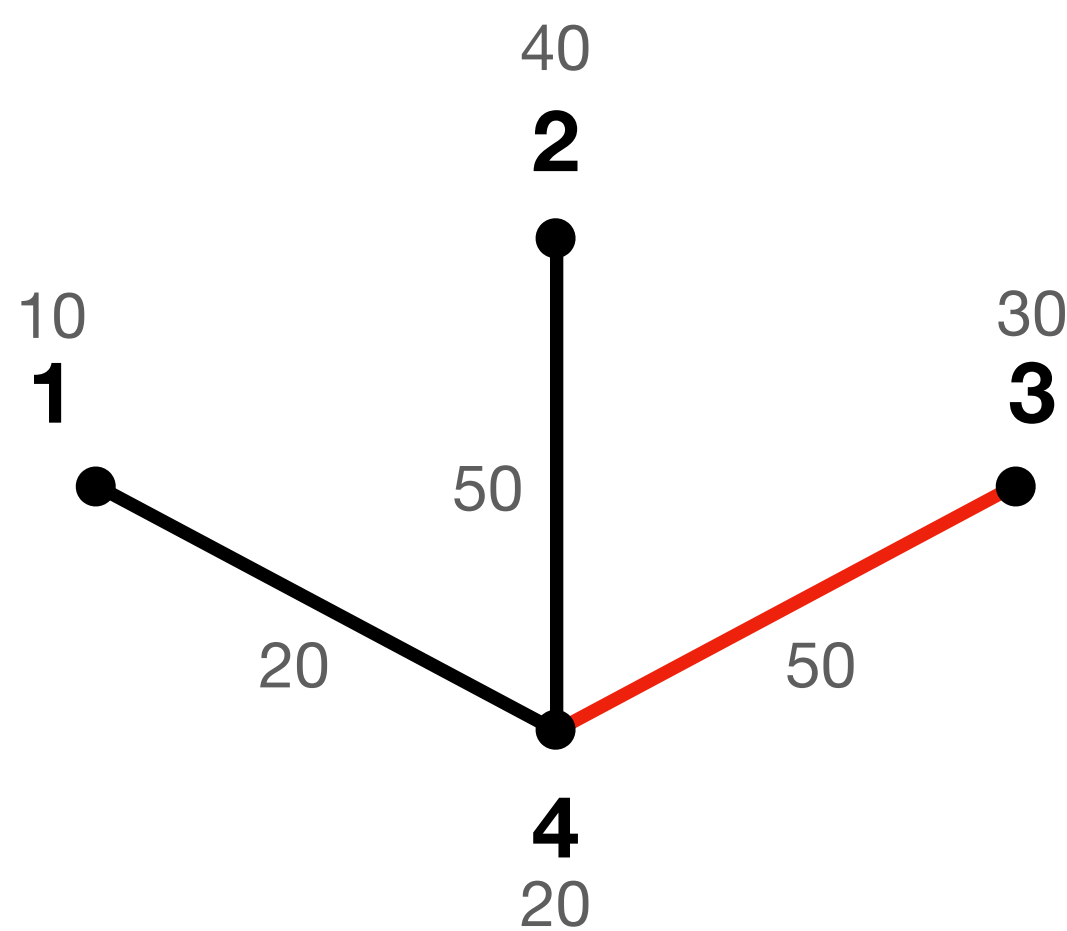
Boundary matrix reduction



	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=50

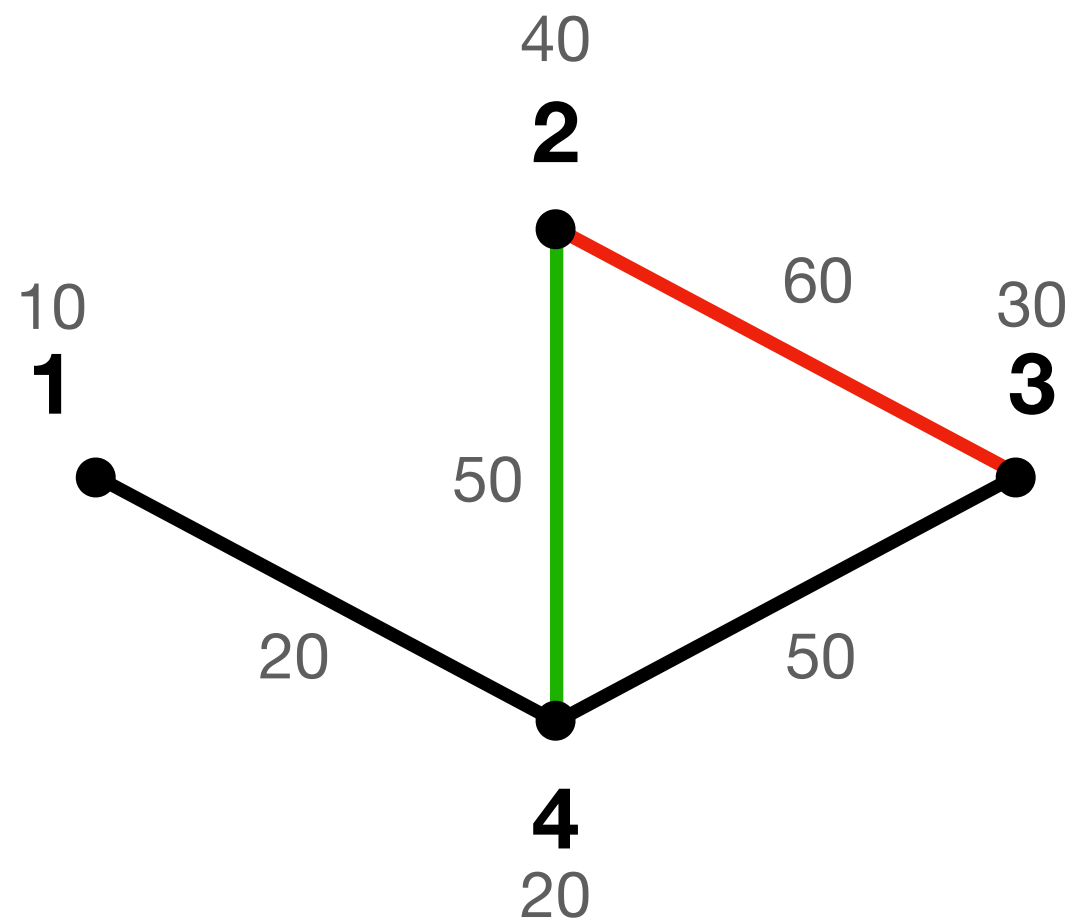
Boundary matrix reduction



	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=50

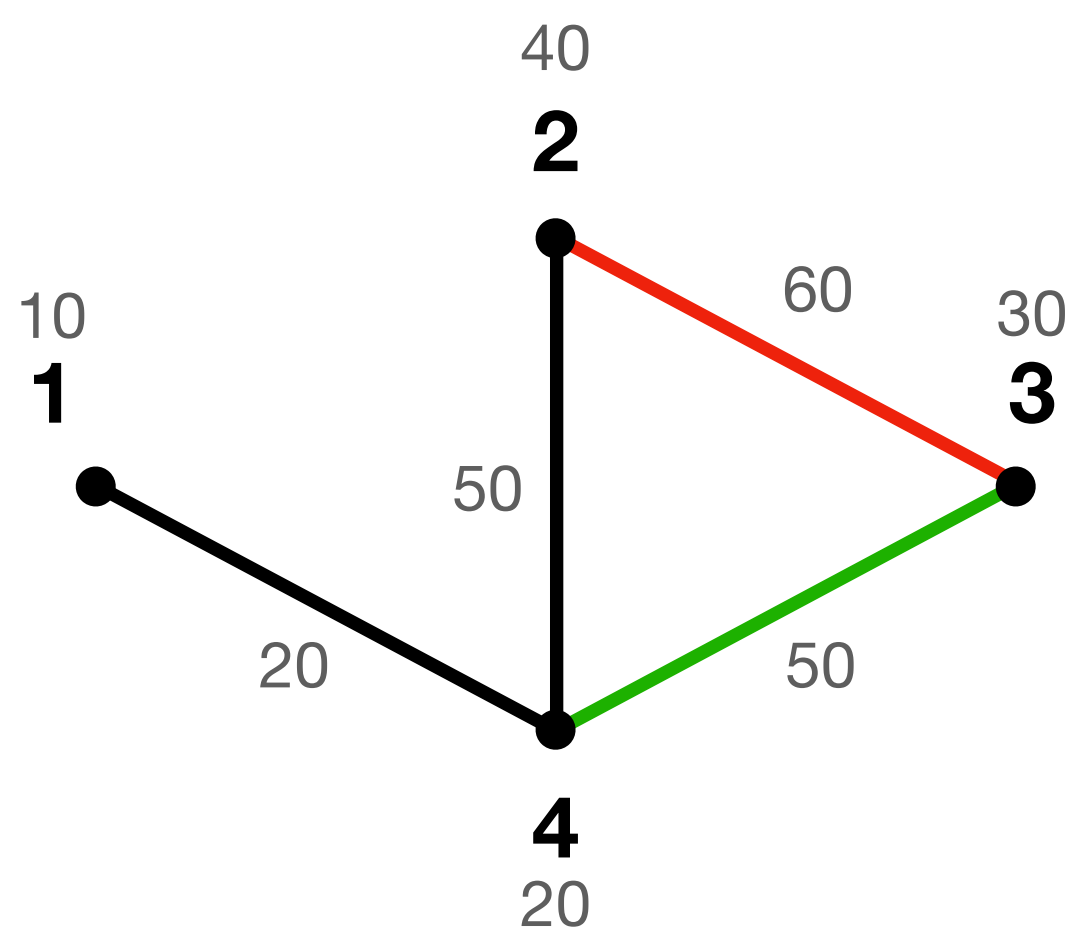
Boundary matrix reduction



	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=60

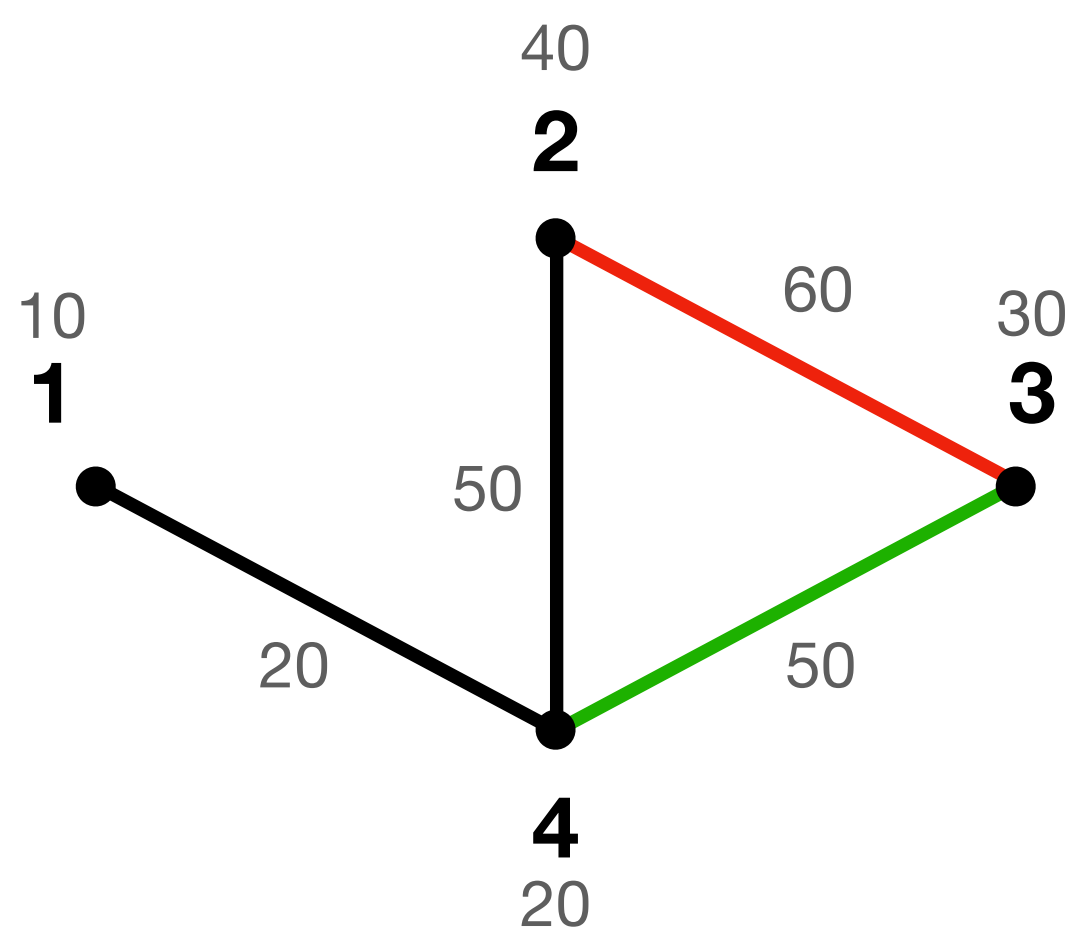
Boundary matrix reduction



	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1	1		
14										
3							1	1		
2						1				1
24									1	
34									1	
23									1	
12										

t=60

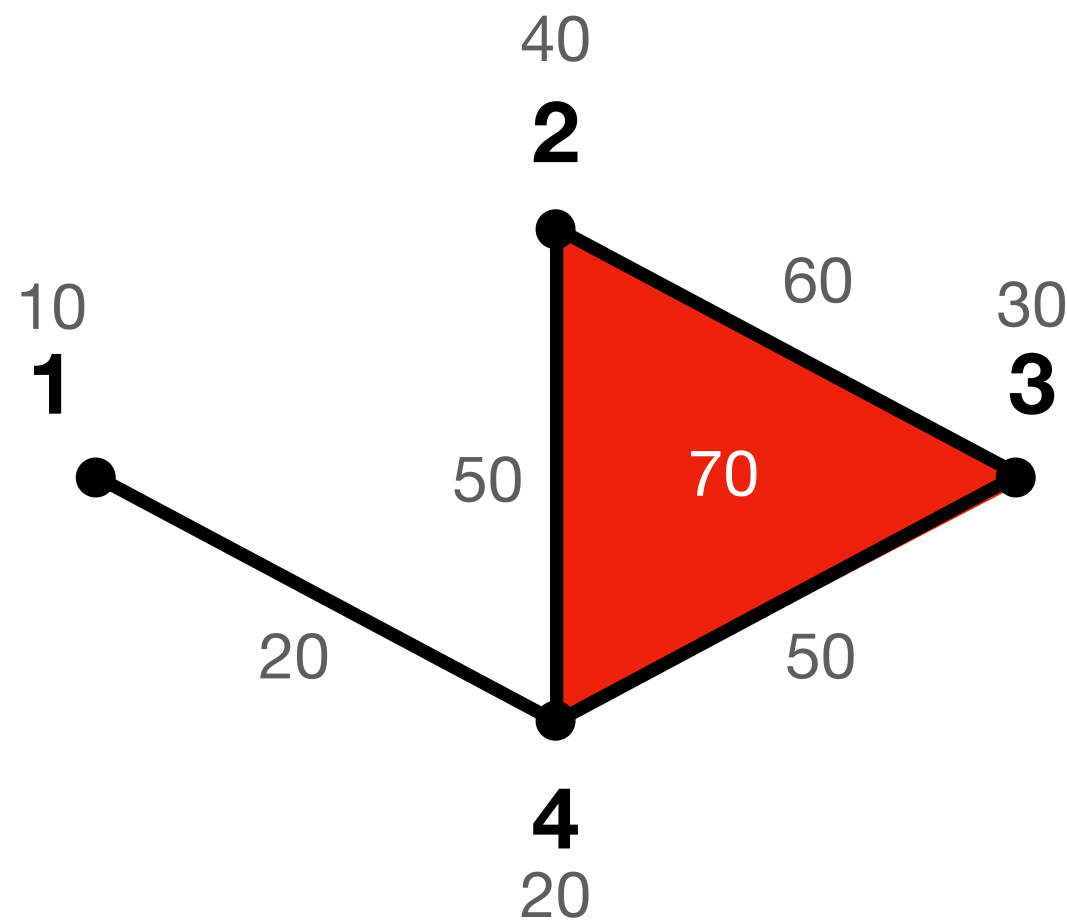
Boundary matrix reduction



	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1			
2						1				1
24									1	
34									1	
23									1	
12										

t=60

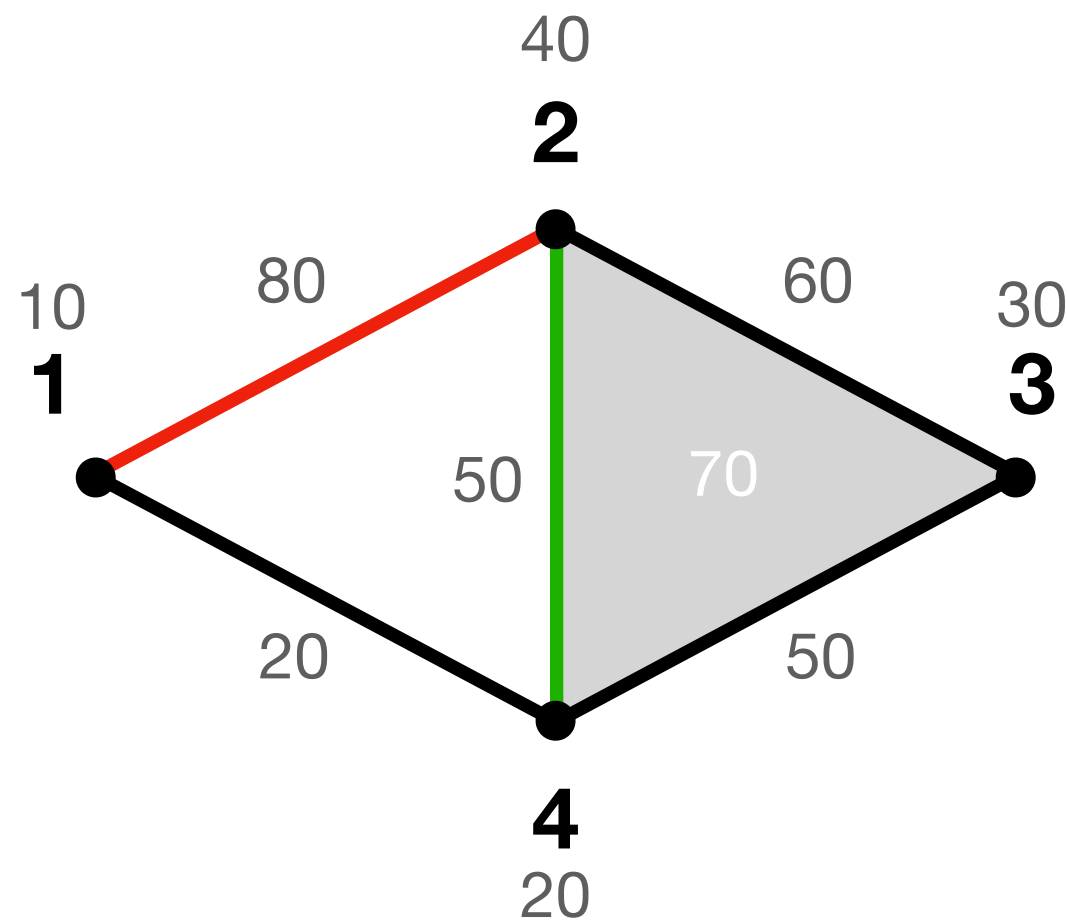
Boundary matrix reduction



	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1			
2						1				1
24									1	
34									1	
23									1	
12										

t=70

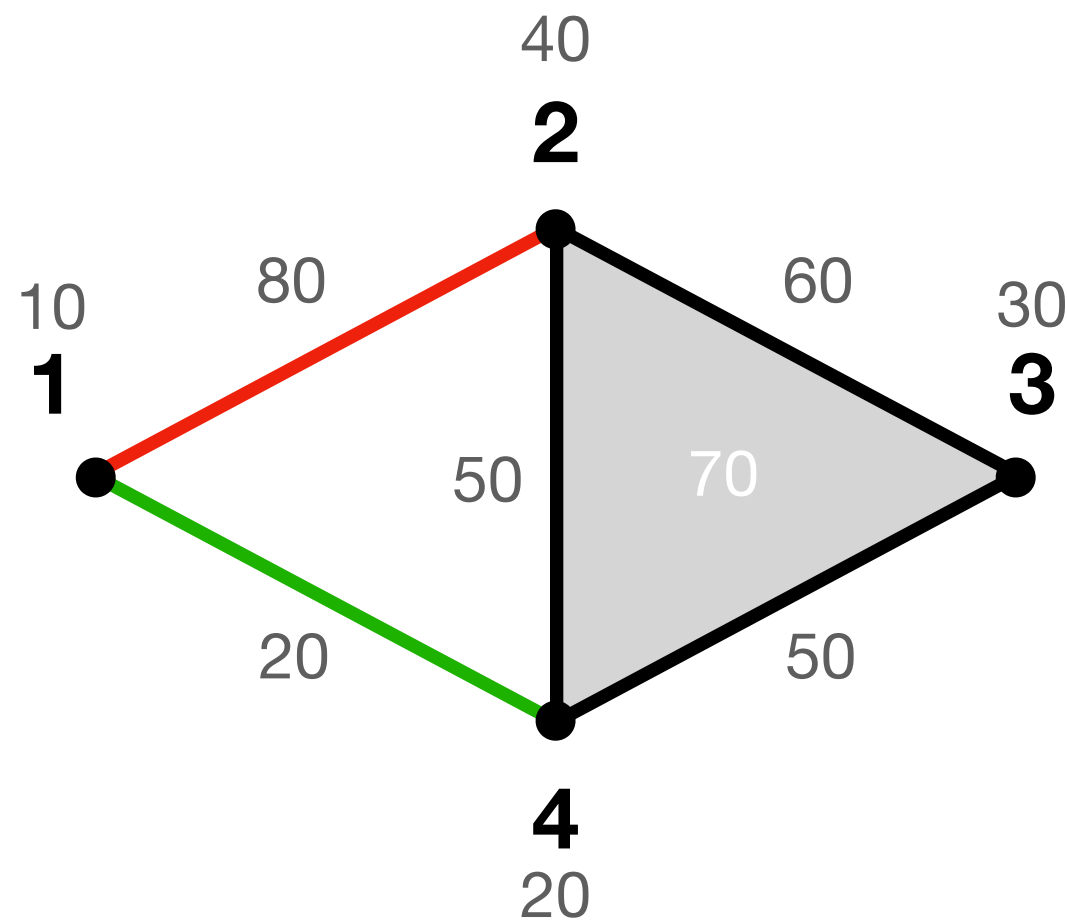
Boundary matrix reduction



	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1			
2						1				1
24									1	
34									1	
23									1	
12										

t=80

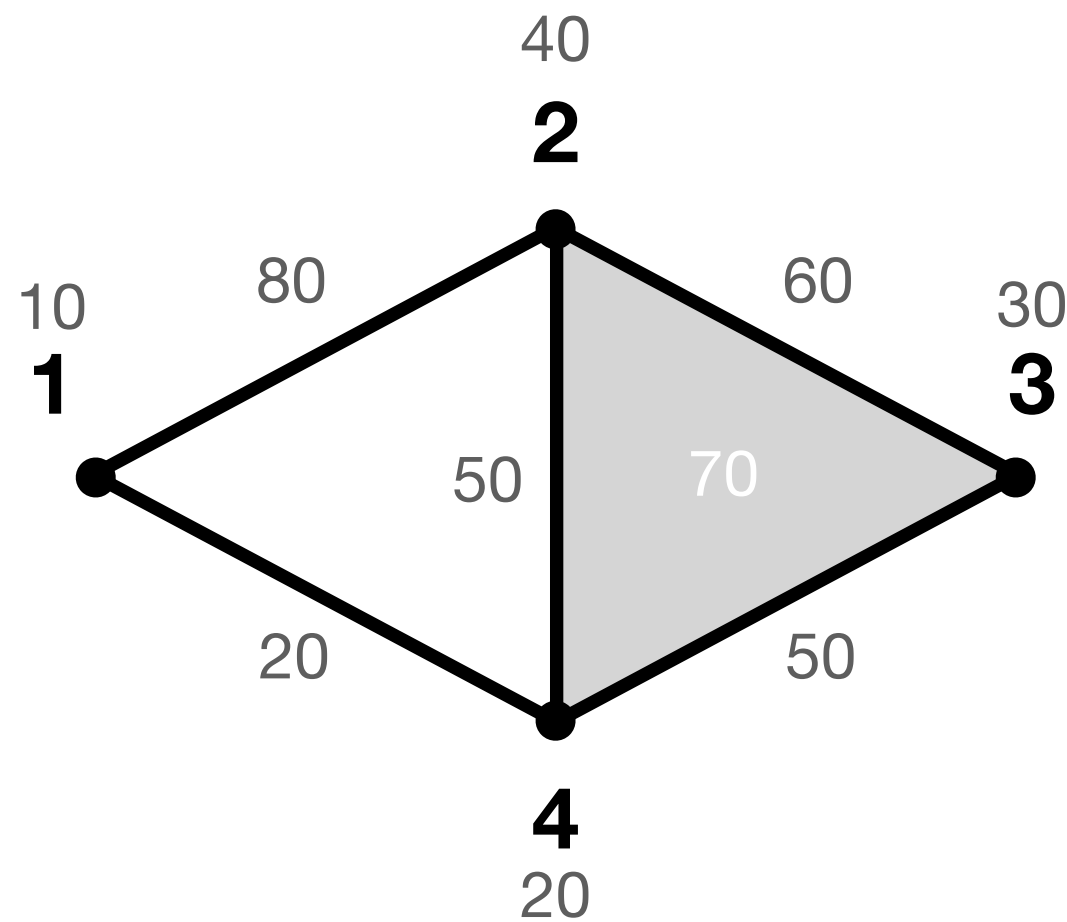
Boundary matrix reduction



	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			1
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

t=80

Boundary matrix reduction

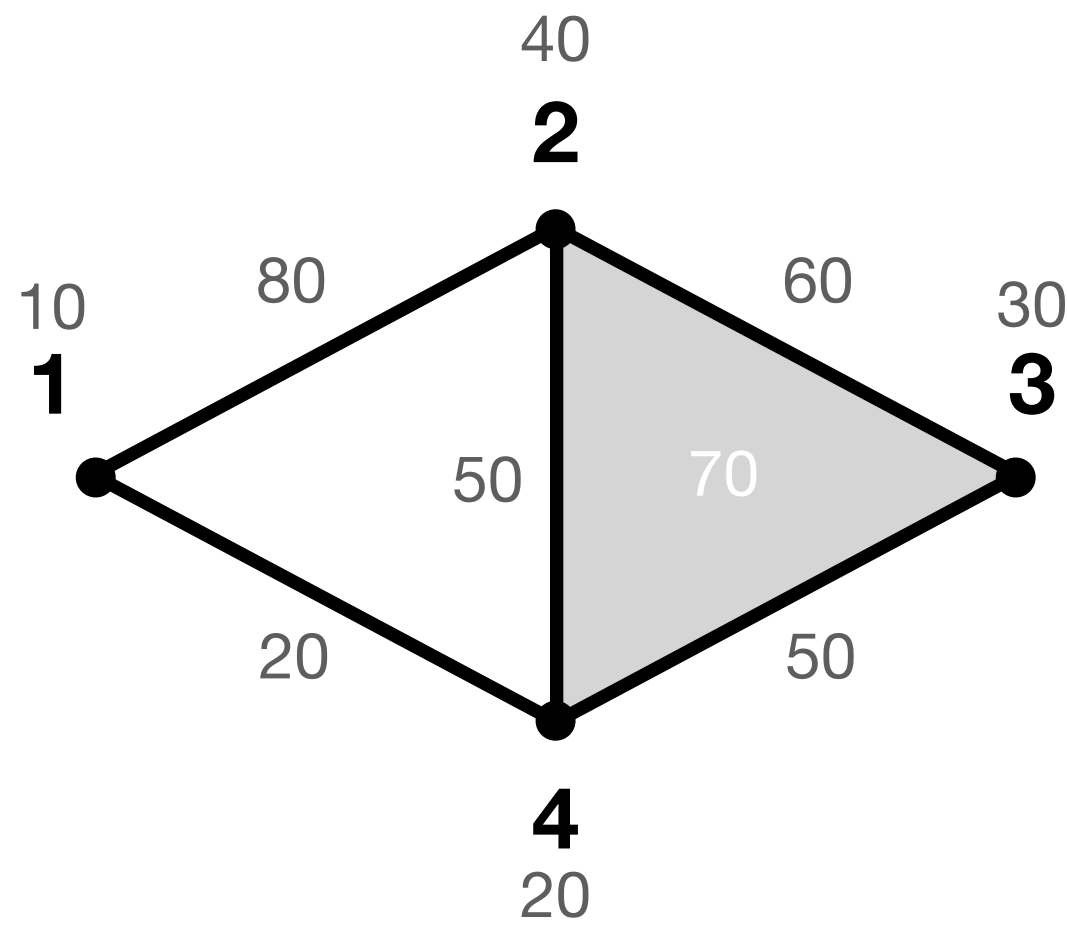


R =

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

Matrix is called reduced if
all lowest nonzero elements are in unique rows

Persistence pairing



R =

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

Essential simplices correspond to unpaired empty columns

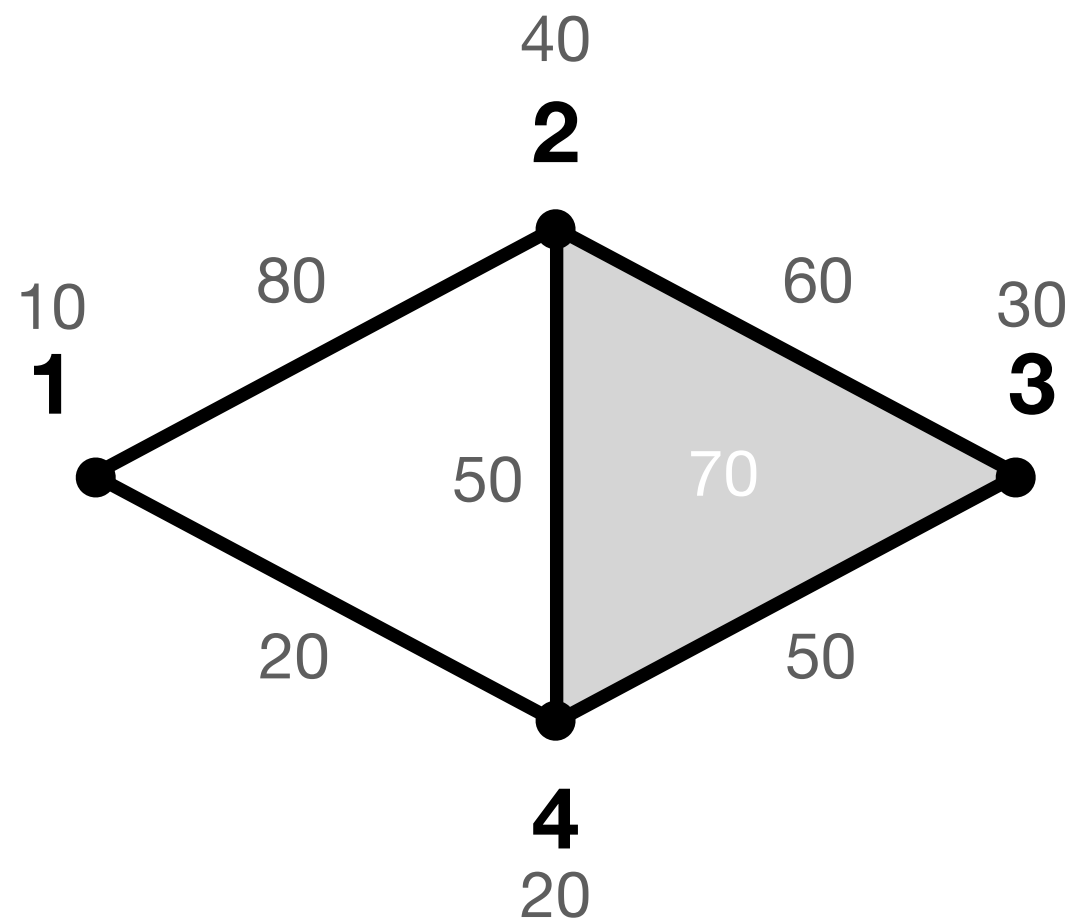
Persistence pairing

<i>P</i>	<i>E</i>
(4, 14) 0	(1, <u>∅</u>) 0
(2, 24) 0	(12, <u>∅</u>) 1
(3, 34) 0	
(23, 234) 1	

$$P = \{(i, j) \mid i = \text{low}(\partial_j), \partial_j \neq 0\}$$

$$E = \{j \mid j \notin \text{low}(\cdot), \partial_j = 0\}$$

Persistence diagram



R =

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

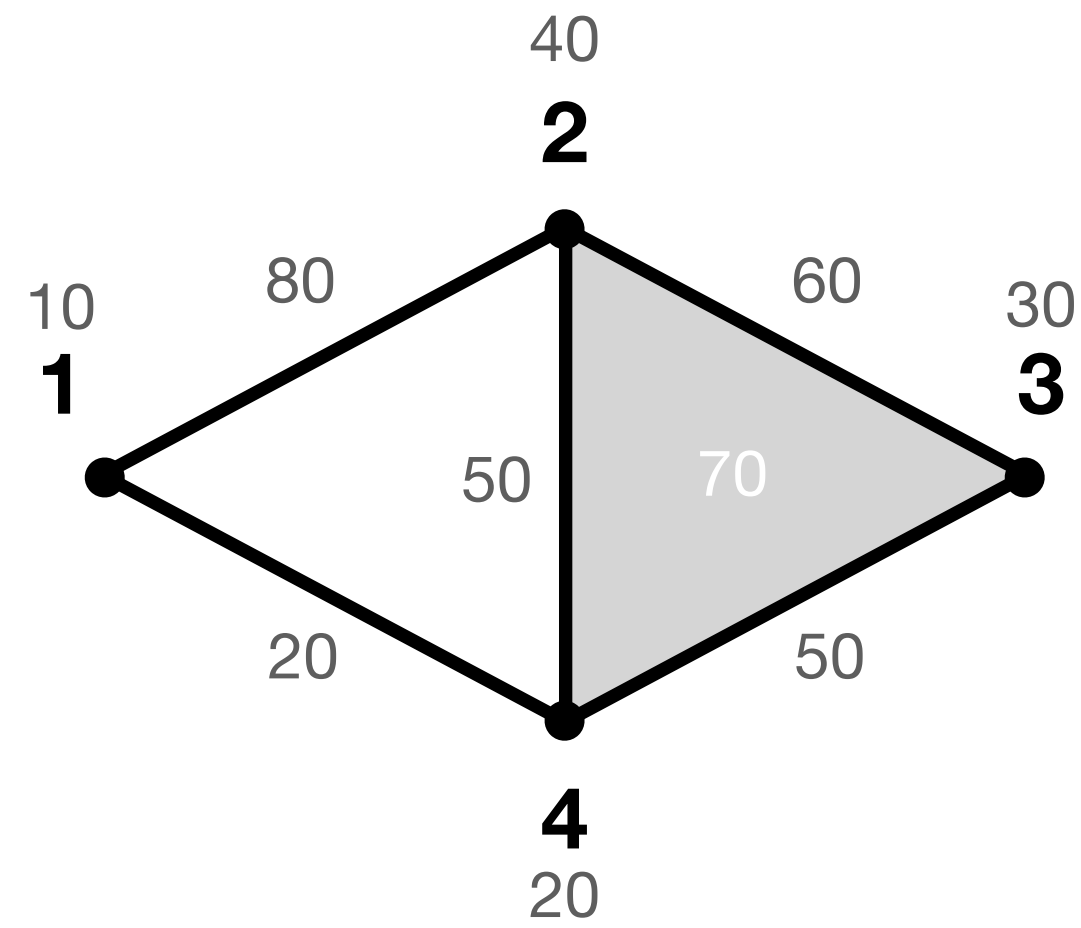
Persistence pairing

<i>P</i>	<i>E</i>
(4, 14) 0	(1, <u>∅</u>) 0
(2, 24) 0	(12, <u>∅</u>) 1
(3, 34) 0	
(23, 234) 1	

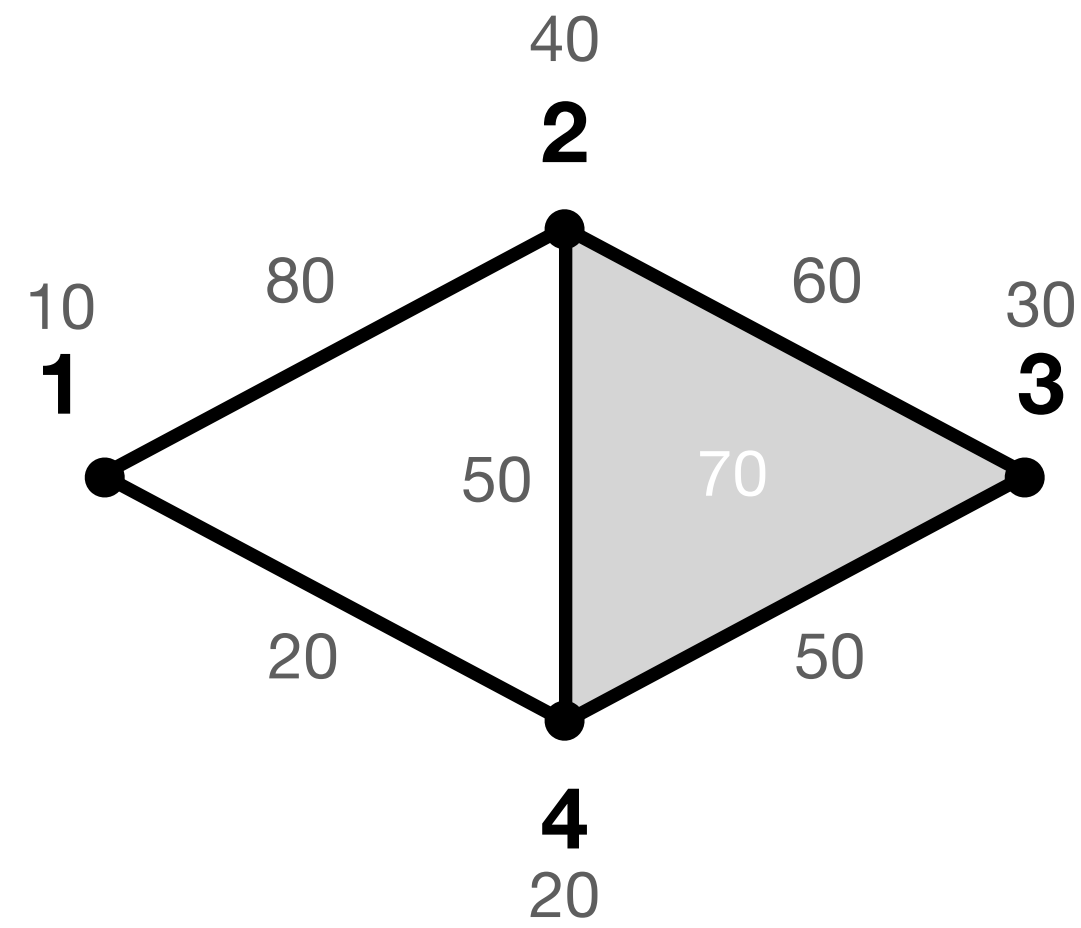
Persistence diagram

(20, 20) 0	(10, ∞) 0
(40, 50) 0	(80, ∞) 1
(30, 50) 0	
(60, 70) 1	

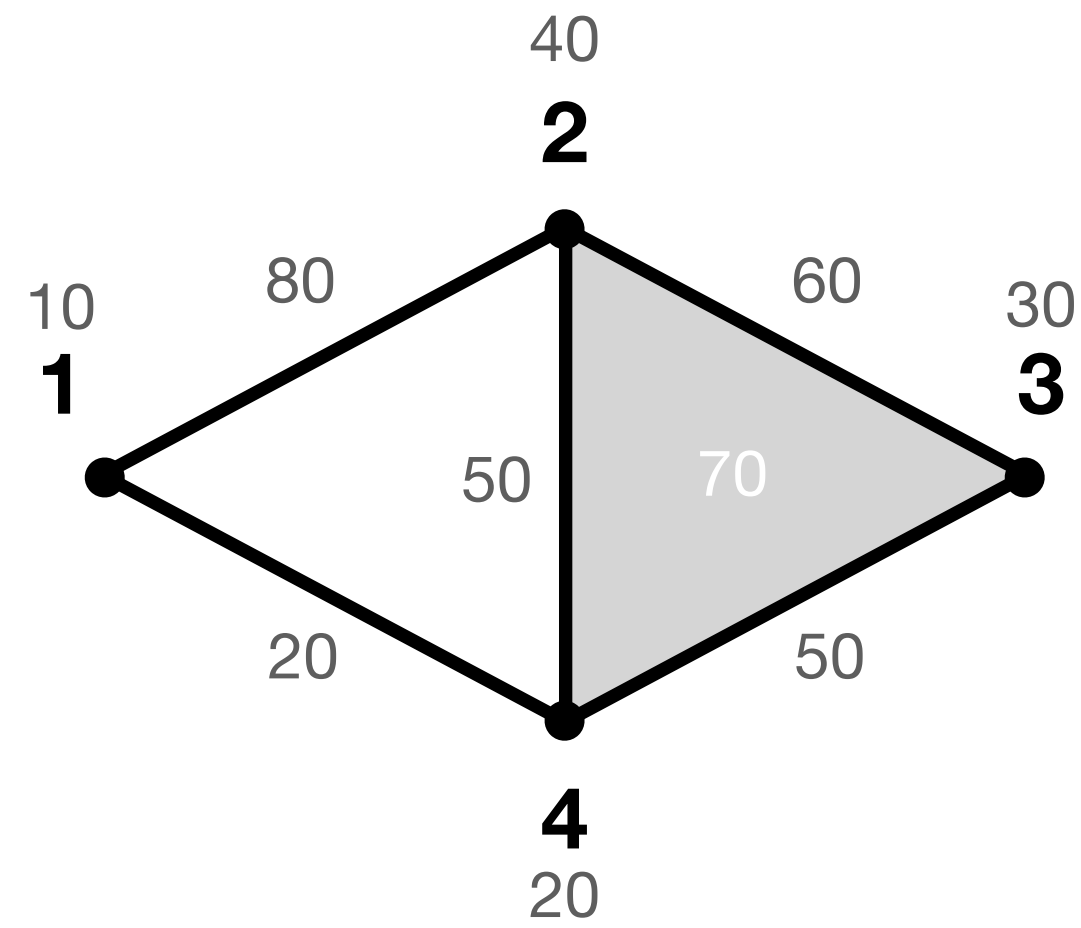
Coboundary matrix reduction

[illegible]

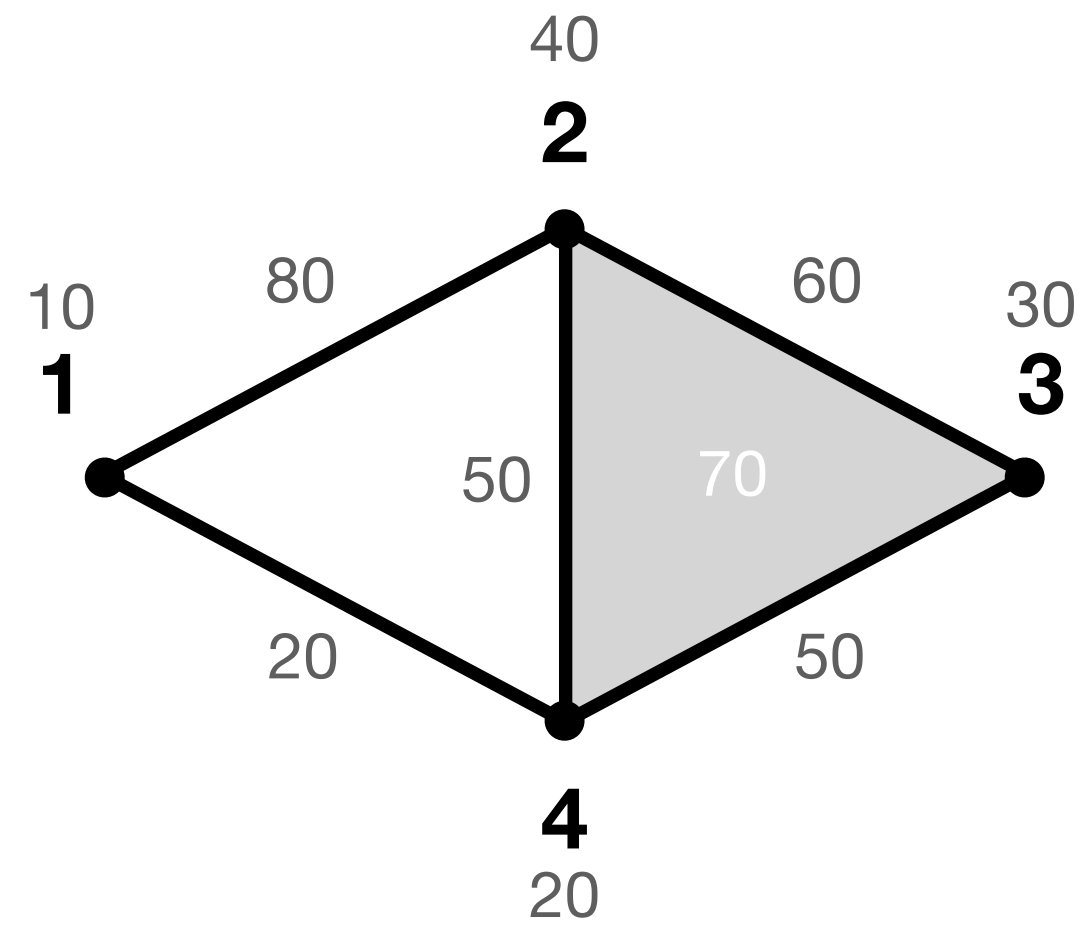
Coboundary matrix reduction

[illegible]

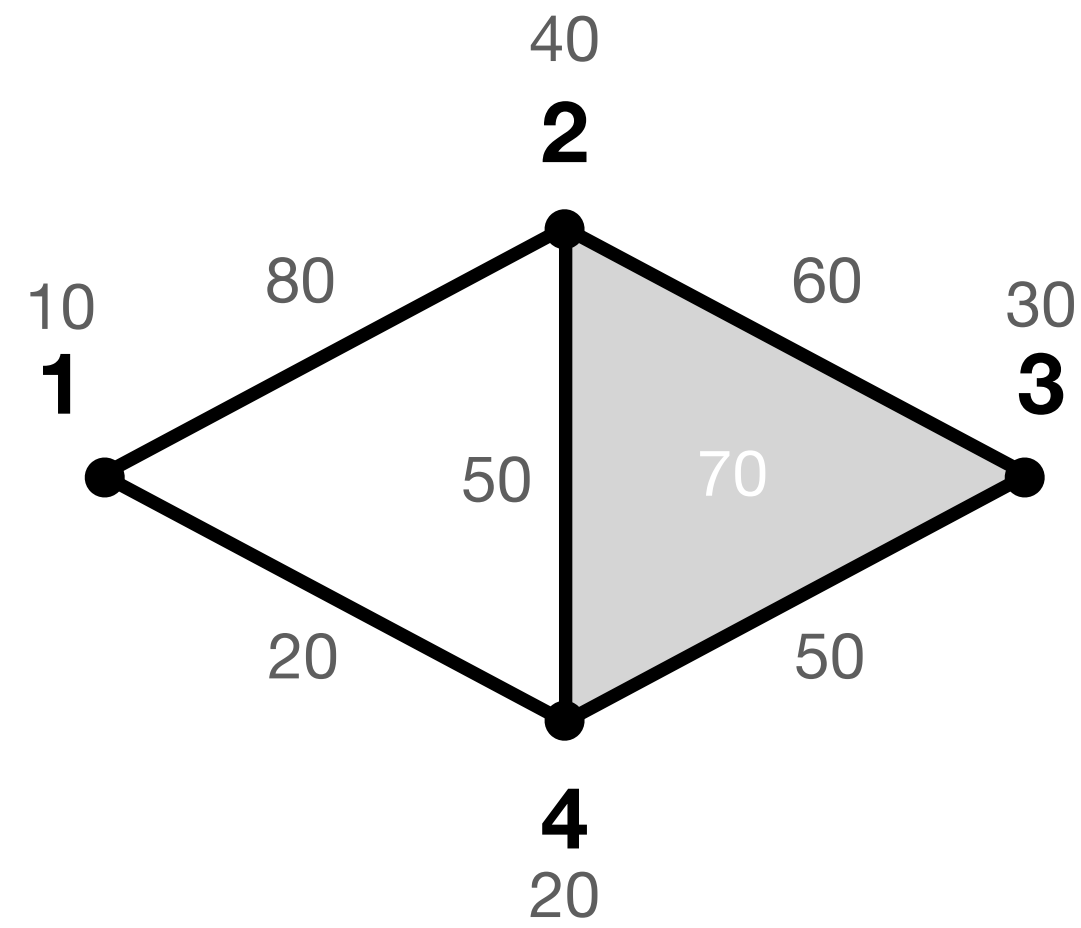
Coboundary matrix reduction

[illegible]

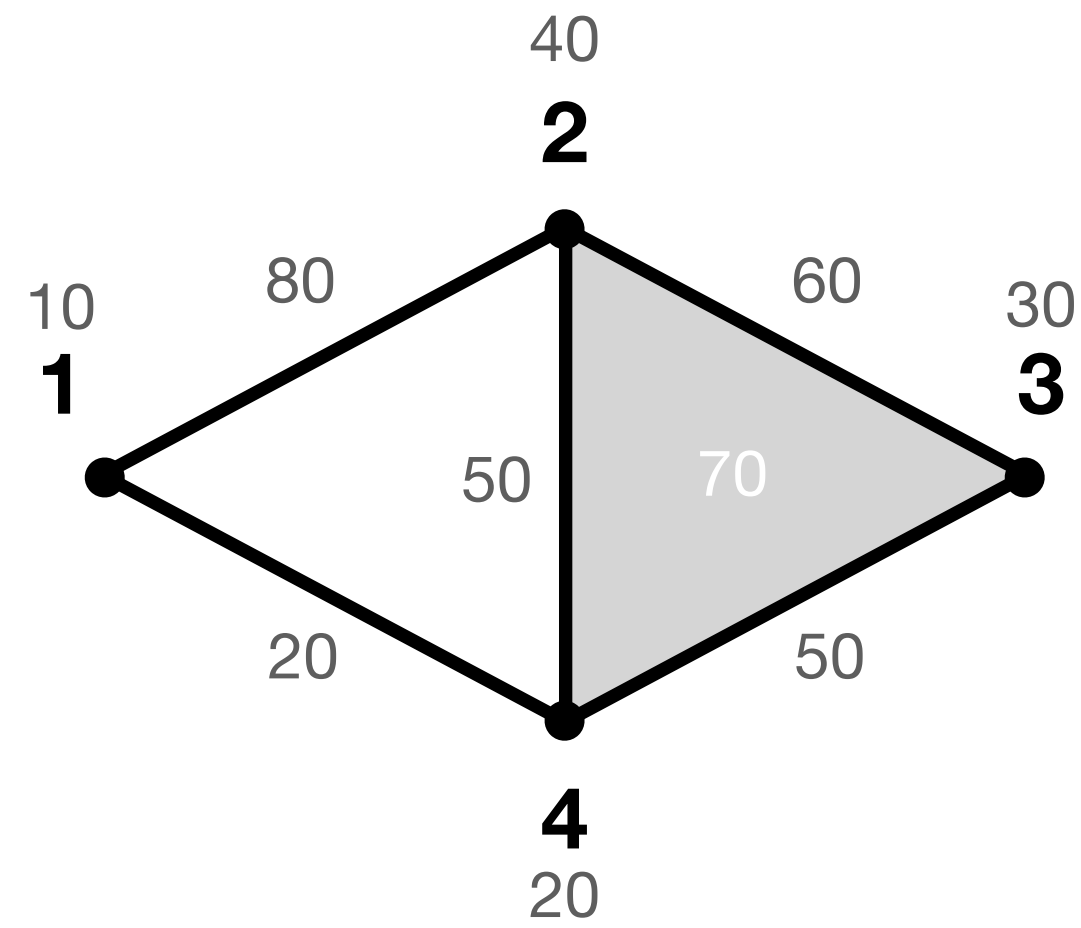
Coboundary matrix reduction

[illegible]

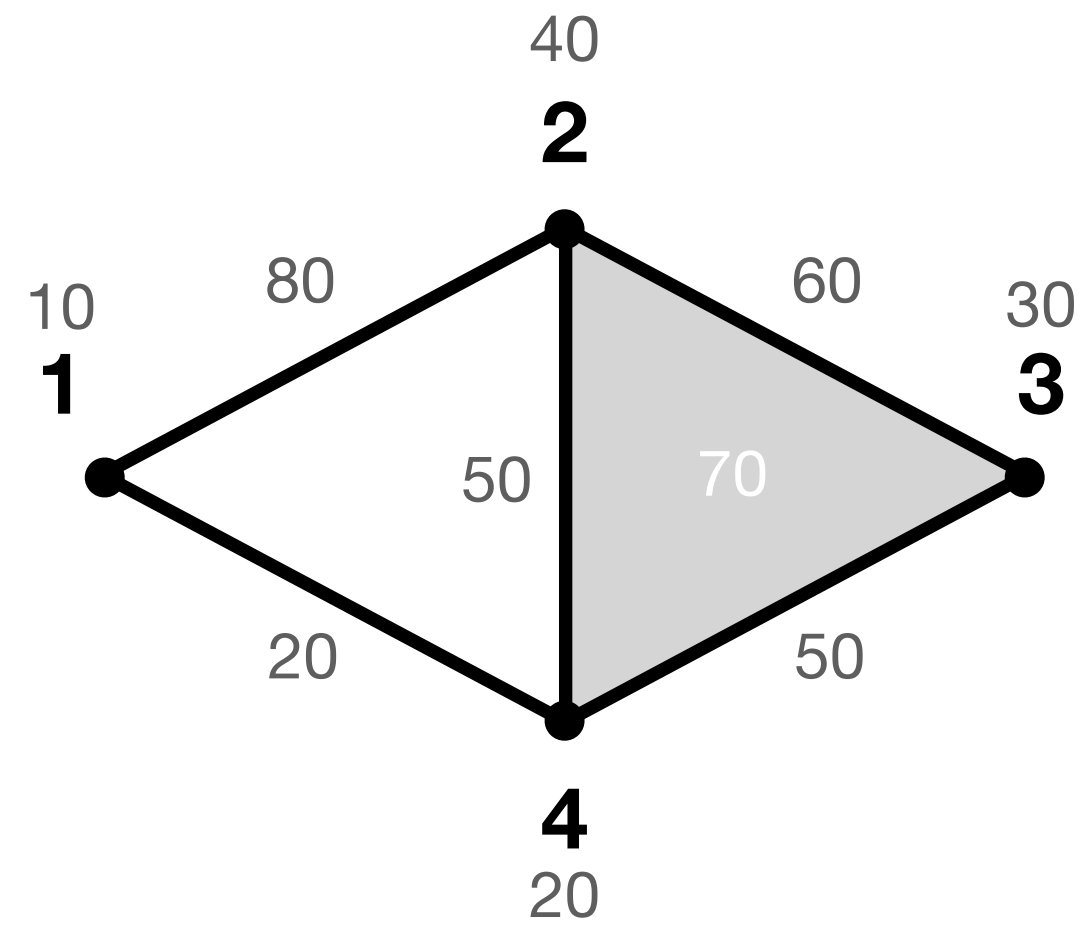
Coboundary matrix reduction

[illegible]

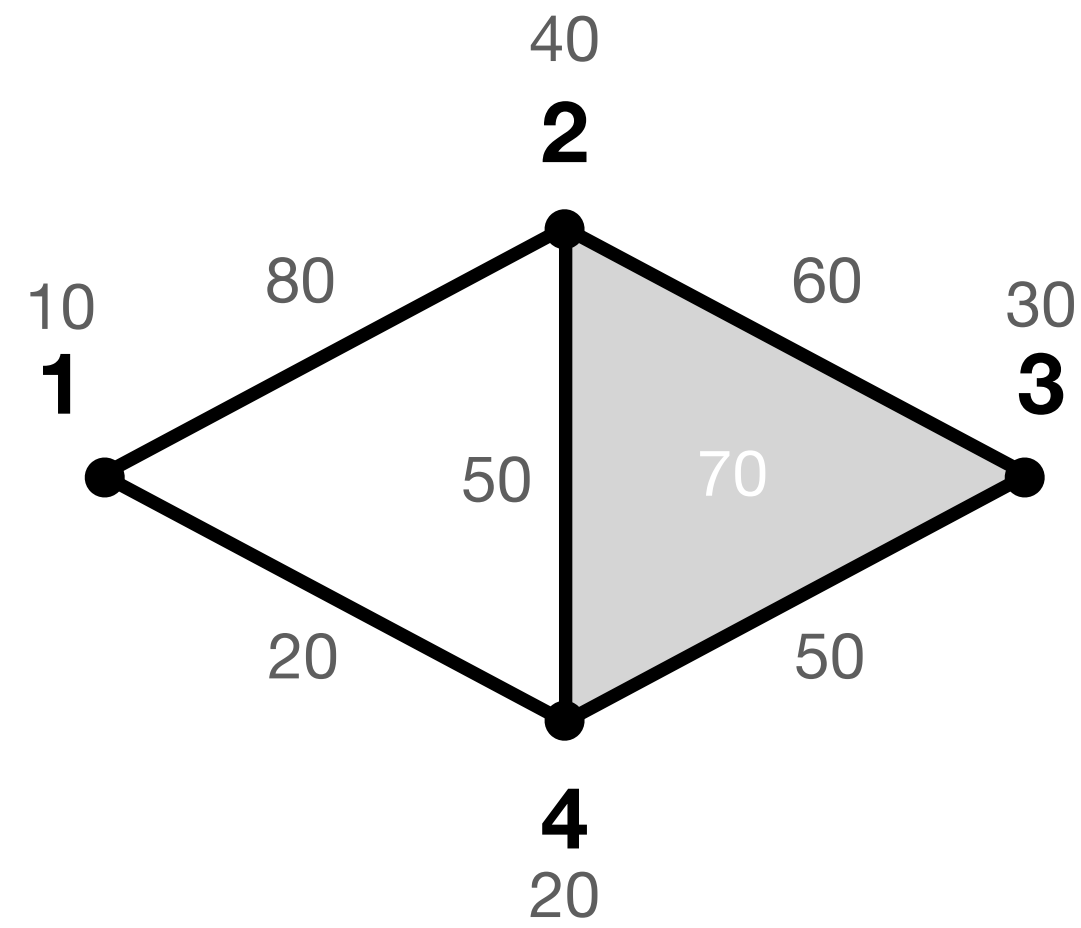
Coboundary matrix reduction

[illegible]

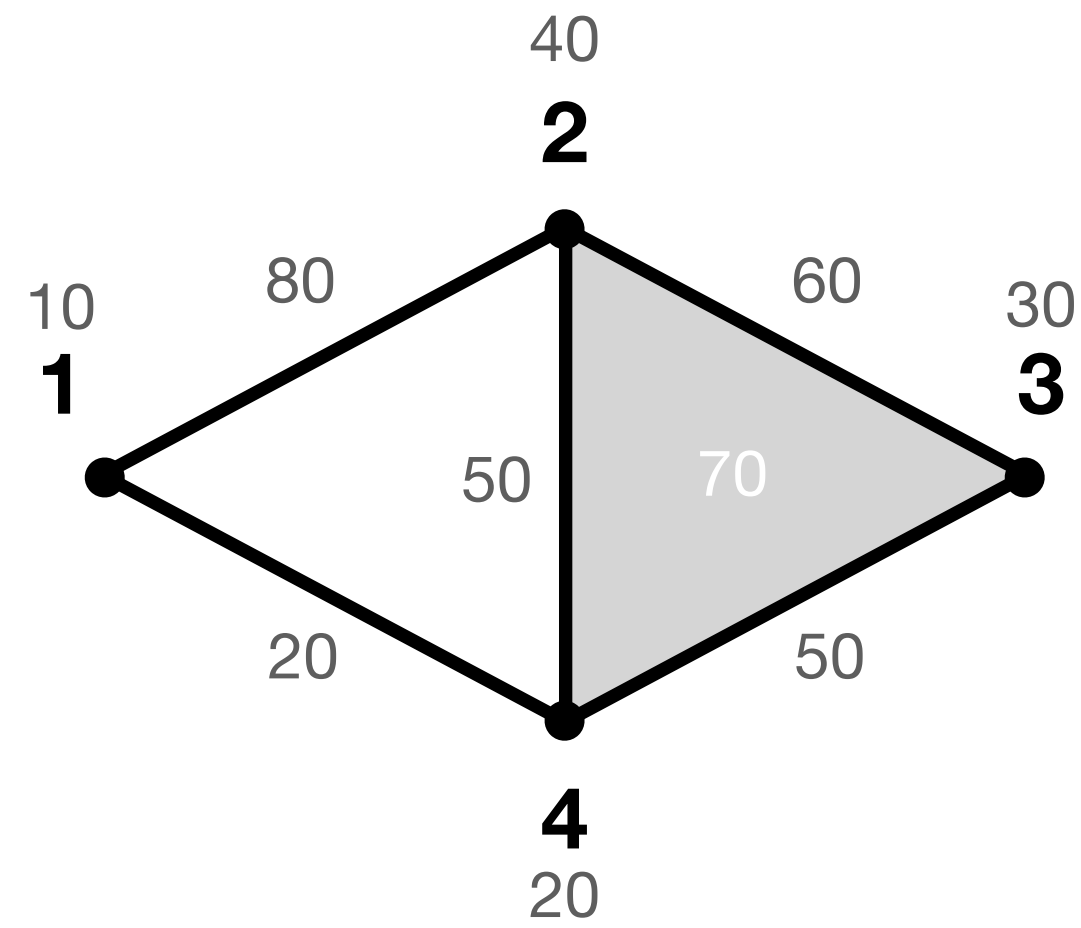
Coboundary matrix reduction

[illegible]

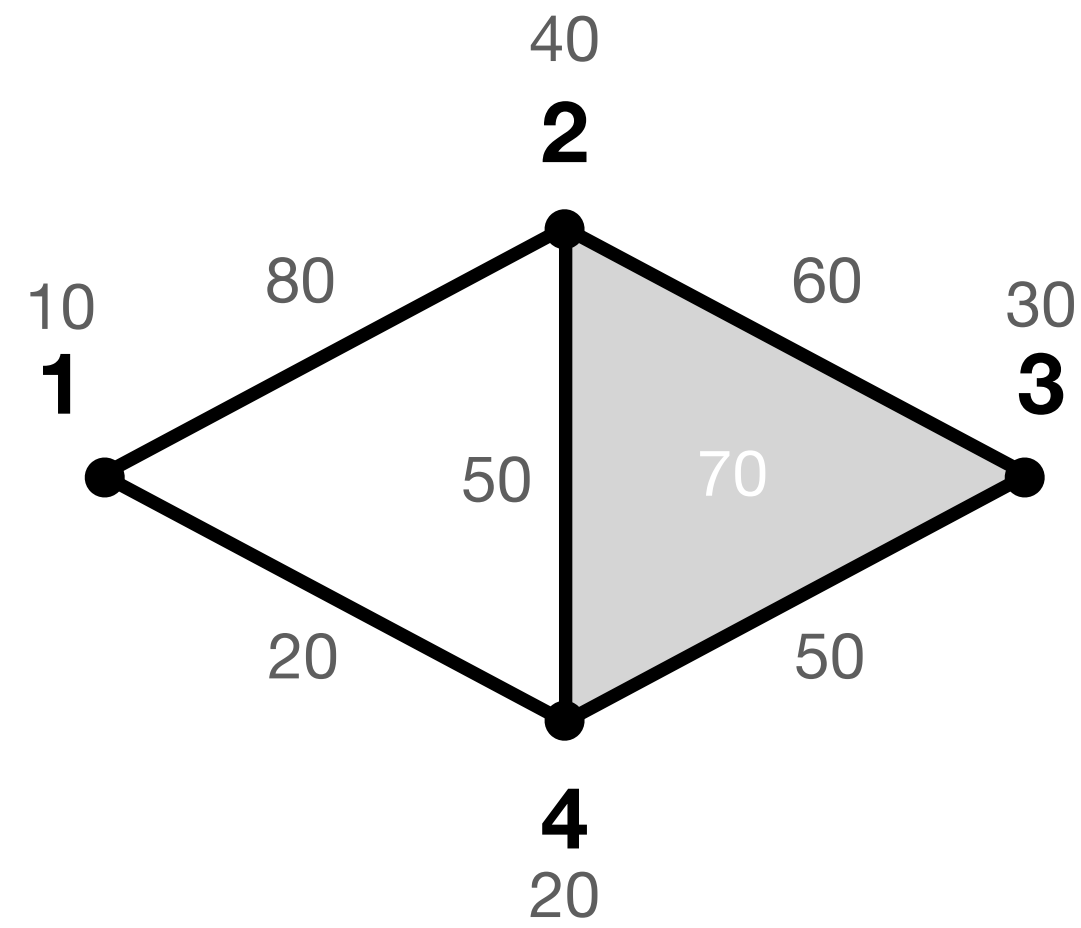
Coboundary matrix reduction

[illegible]

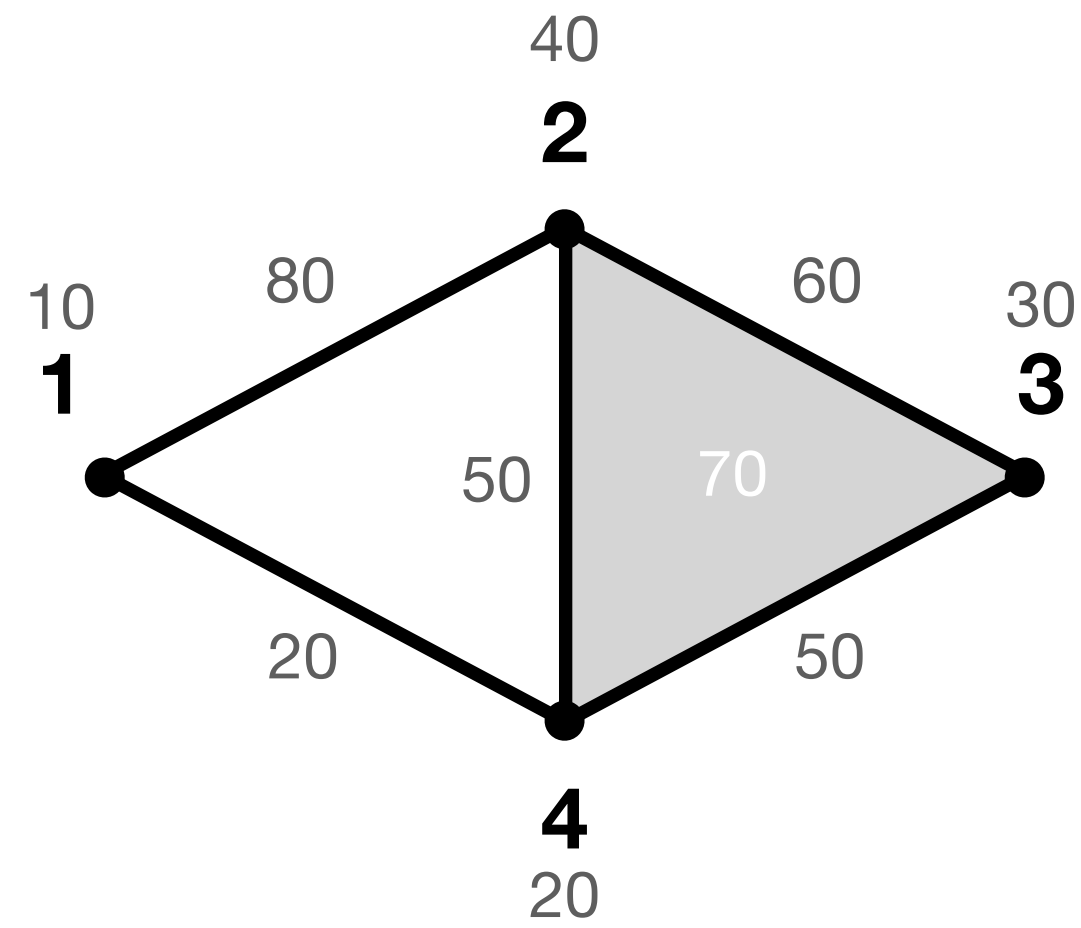
Coboundary matrix reduction

[illegible]

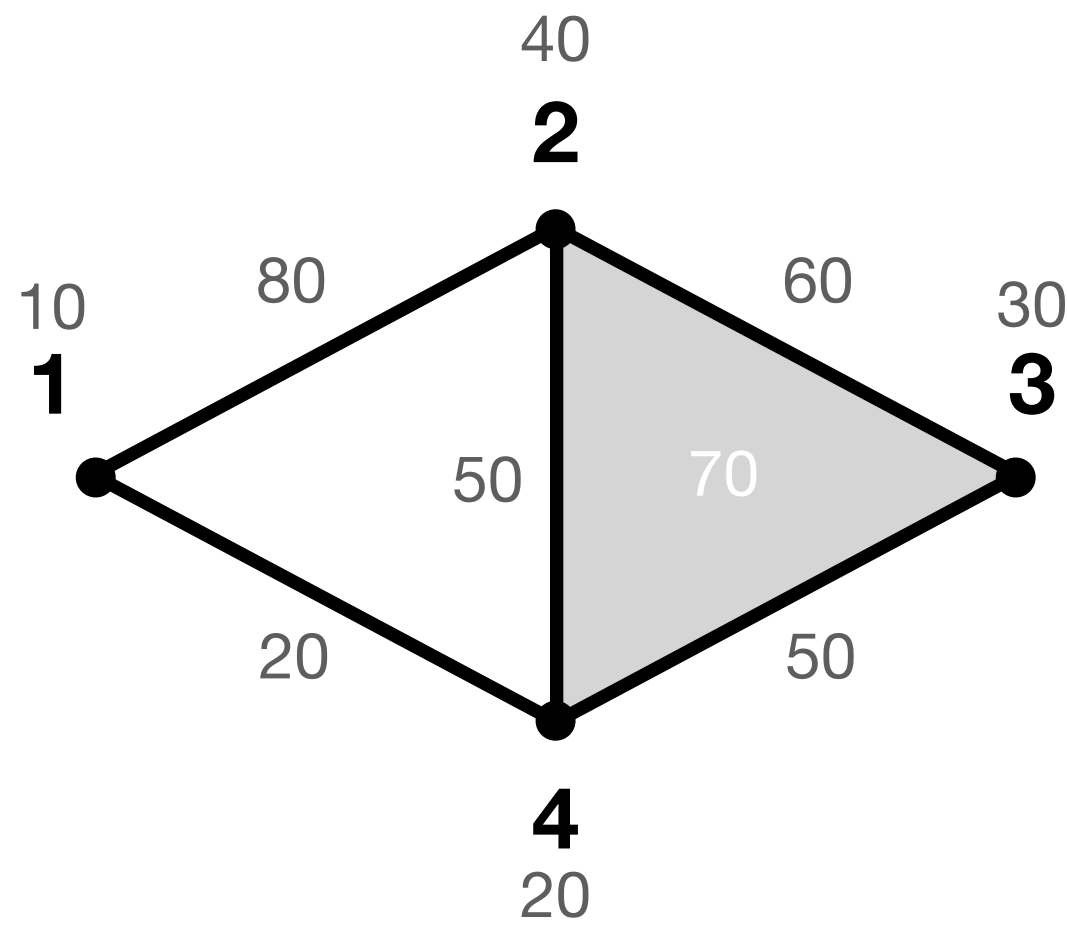
Coboundary matrix reduction

[illegible]

Coboundary matrix reduction

[illegible]

Persistence pairing



	80	70	60	50	50	40	30	20	20	10
	12	234	23	34	24	2	3	14	4	1
12						1				
234			1							
23						1	1			
34							1		1	
24						1			1	
2										
3										
14									1	
4										
1										

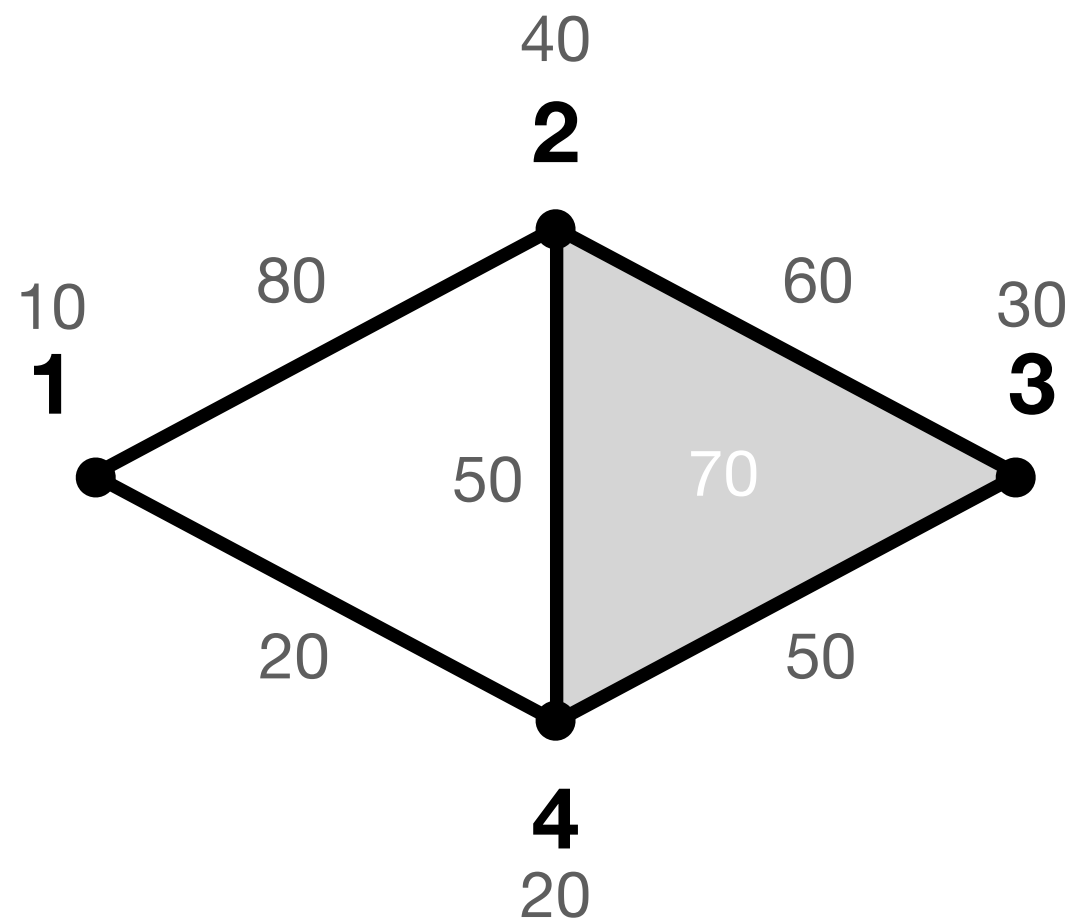
Persistence pairing

<i>P</i>	<i>E</i>
(4, 14) 0	(1, <u>∅</u>) 0
(2, 24) 0	(12, <u>∅</u>) 1
(3, 34) 0	
(23, 234) 1	

$$P = \{(j, i) \mid i = \text{low}(\partial_j), \partial_j \neq 0\}$$

$$E = \{j \mid j \notin \text{low}(\cdot), \partial_j = 0\}$$

Persistence diagram



	80	70	60	50	50	40	30	20	20	10
	12	234	23	34	24	2	3	14	4	1
12						1				
234			1							
23						1	1			
34							1		1	
24						1			1	
2										
3										
14									1	
4										
1										

Persistence pairing

P	E
(4, 14) 0	(1, $\emptyset$) 0
(2, 24) 0	(12, $\emptyset$) 1
(3, 34) 0	
(23, 234) 1	

Persistence diagram

(20, 20) 0	(10, ∞) 0
(40, 50) 0	(80, ∞) 1
(30, 50) 0	
(60, 70) 1	

Clearing optimization

The reduction of a column that corresponds to a negative simplex (one that destroys an homology class) also reveals the simplex that creates the corresponding class.

Since that column is known to be zero after reduction, we can immediately set it to zero.

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

Clearing optimization

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

Clearing optimization

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1			
2						1				1
24									1	
34									1	
23									1	
12										

Clearing optimization

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			1
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

Clearing optimization

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

Parallel algorithm

Given a matrix ∂ , the leftmost columns are defined

$$L := \{\partial_j \mid \nexists k < j : \text{low}(k) = \text{low}(j) > -1\}$$

Given a leftmost column $\ell \in L$, its neighbors

$$N(\ell) := \{j > \ell \mid \text{low}(j) = \text{low}(\ell)\}$$

Algorithm 4: Parallel column additions

Input: An $n \times n$ boundary matrix ∂ over $\mathbb{Z}_2$

```
1 while  $\partial$  is not reduced do
2    $\mathcal{L} \leftarrow \{j \mid \nexists k < j : \text{low}_\partial(k) = \text{low}_\partial(j) > -1\}$ 
3   for  $\ell \in \mathcal{L}$  do
4     for  $j \in \mathcal{N}(\ell)$  do
5        $\partial_j \leftarrow \partial_j + \partial_\ell$ 
6     end for
7   end for
8 end while
9 return  $\partial$ 
```

Parallel algorithm

		10	20	20	30	40	50	50	60	70	80
		1	4	14	3	2	24	34	23	234	12
1				1							1
4				1			1	1			
14											
3								1	1		
2							1		1		1
24										1	
34										1	
23										1	
12											

Leftmost, neighbors

Parallel algorithm

	10	20	20	30	40	50	50	60	70	80	
	1	4	14	3	2	24	34	23	234	12	Leftmost, neighbors
1			1							1	
4			1			1	1	1		1	
14											
3							1	1			
2						1					
24									1		
34									1		
23									1		
12											

Parallel algorithm

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1	1		1
14										
3							1	1		
2						1				
24									1	
34									1	
23									1	
12										

Leftmost, neighbors

Leftmost, neighbors

Parallel algorithm

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

Leftmost, neighbors

Leftmost, neighbors

Parallel algorithm with clearing optimization

Given a matrix ∂ , the leftmost columns are defined

$$L := \{\partial_j \mid \nexists k < j : \text{low}(k) = \text{low}(j) > -1\}$$

Given a leftmost column $\ell \in L$, its neighbors

$$N(\ell) := \{j > \ell \mid \text{low}(j) = \text{low}(\ell)\}$$

Algorithm 5: The final form of our implemented algorithm

Input: An $n \times n$ boundary matrix ∂ over $\mathbb{F}_2$

```
1 while  $\partial$  is not reduced do
2   for  $j = 0, \dots, n - 1$  do
3     | Set column  $\text{low}_\partial(j)$  to zero
4   end for
5    $\mathcal{L} \leftarrow \{j \mid \nexists k < j : \text{low}_\partial(k) = \text{low}_\partial(j) > -1\}$ 
6   for  $\ell \in \mathcal{L}$  do
7     for  $j \in \mathcal{N}(\ell)$  do
8       |  $\partial_j \leftarrow \partial_j + \partial_\ell$ 
9     end for
10  end for
11 end while
12 return  $\partial$ 
```

Parallel algorithm with clearing optimization

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

Parallel algorithm with clearing optimization

	10	20	20	30	40	50	50	60	70	80	
	1	4	14	3	2	24	34	23	234	12	Leftmost, neighbors
1			1							1	
4			1			1	1				
14											
3							1				
2						1				1	
24									1		
34									1		
23									1		
12											

Parallel algorithm with clearing optimization

	10	20	20	30	40	50	50	60	70	80	
	1	4	14	3	2	24	34	23	234	12	Leftmost, neighbors
1			1							1	
4			1			1	1			1	
14											
3							1				
2						1					
24									1		
34									1		
23									1		
12											

Parallel algorithm with clearing optimization

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										